

EBC4097

Quantitative Techniques for Financial Economics: A Study Book

A friendly study book for beginners — built around the lecture slides and the sample exam

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All examples use the course datasets. Notation follows the lecture slides and the sample exam.

Read this first

This book is written for someone who has never studied time series before. It assumes you remember what a mean, a variance and a regression are — nothing more. Every idea is introduced with a picture in words before any formula appears, and every formula is followed by a sentence saying what it means. If you read it from start to finish, slowly, you will understand everything the exam asks.

What the exam looks like

The sample exam tells us exactly what to prepare for:

- **Closed book**, but you may bring **two A4 sheets of handwritten notes** (both sides). Chapter 11 gives you a ready-made formula sheet to copy onto them.
- **Four questions, 100 points, 55 to pass**. Calculators are allowed.
- The four questions in the sample exam cover: **volatility models** (30 points), **a regression with lags, trends and dummies** (40 points), **high-dimensional data: Ridge and PCA** (15 points), and **VAR models** (15 points).
- The instructions say: *answer each question with a full sentence and explain how you obtain your results; clarify any assumptions you make*. So a bare number is never a full answer. Throughout this book, the “Exam corner” boxes show you how to phrase answers.

The types of exam question

Almost every exam sub-question is one of five kinds. Learn to recognise them:

1. **Explain why** — “What is the reason for using model A instead of model B?”
2. **Derive** — “Derive the unconditional mean/variance of ...” (always the same few steps; Chapters 4 and 8 drill them).
3. **Extend the model** — “Add a lunch effect / a weekend effect / volatility in the mean. Which parameter captures it?” (Chapter 6 is entirely about this.)
4. **Spot the problem** — “Why do these parameter values cause a problem?”
5. **Read a picture or a number** — a scree plot, a time series plot, a set of VAR estimates.

How this book is organised

Chapters 1–3 build the vocabulary: what time series are, what “stationary” means, and how we measure a series’ memory. Chapters 4–6 study one series at a time: ARMA models, trends and unit roots, and how to extend a regression with dummies. Chapter 7 handles several series together (VAR). Chapter 8 is about volatility (ARCH, GARCH and friends) — the biggest exam

question. Chapter 9 covers big datasets (factor models, PCA, Ridge, Lasso). Chapter 10 solves the complete sample exam, and Chapter 11 is your formula sheet and final checklist.

Four kinds of boxes help you along: **blue** boxes give the idea in plain words, **green** boxes show it at work in a **real-world example** (many use the actual data from your tutorials), **orange** boxes tell you what the exam wants, and **red** boxes warn about the mistakes students make most often.

Three appendices complete the syllabus: Appendix A collects the R commands and the lessons from the tutorials, Appendix B summarises the group assignment, and Appendix C is a **coverage map** linking every syllabus session and every lecture-slide topic to the place in this book where it is explained.

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Chapter 1

What Is a Time Series, and Why Is It Special?

1.1 Data that arrive in order

A **time series** is a list of observations of the same thing, recorded one after another in time: Tesla's closing price every trading day, euro-area inflation every month, the 10-year US government yield every week. We write it as $y_1, y_2, \dots, y_T$, where t is the time index and T is the number of observations.

In your earlier econometrics courses you probably worked with *cross-section* data: many different people or firms observed at one moment. Time series data differ in one crucial way:

The idea in plain words

In a time series, **the order matters, and neighbouring observations are related**. Today's interest rate is very close to yesterday's. That dependence over time is the most distinctive feature of time series data — and it is exactly what lets us **forecast**, which is usually the goal.

The lecture asks a simple question about the US 10-year yield: *does its history help predict its future?* That question — “does the past tell us something about the future?” — runs through the whole course.

1.2 Stock and flow data

The slides make a small but useful distinction about *what* the time stamp t means.

- For a **stock** variable (a level measured at a moment), t is the exact moment of measurement: Tesla's closing price at 16:00 on 1 April 2020, or the amount of money in circulation on 31 December.
- For a **flow** variable (an amount accumulated over a period), t is the period: GDP in the first quarter of 2025, sales in January, the number of trades between 10:00 and 10:05.

1.3 What do we use time series for?

The lecture names three uses. All three need not only a number but also a sense of how *uncertain* that number is.

1. **Measuring economic relations.** How strongly does money demand respond to GDP (its *elasticity*)? Tools: t -tests for one coefficient, F -tests for several.
2. **Policy analysis.** A central bank wants to know how money demand responds to the interest rate it controls, so it can predict what a rate change will do.

3. **Forecasting.** What will GDP growth be next year? We can give a single best guess (a *point forecast*), a likely range (an *interval*, e.g. 95%), or even a full probability distribution.

Real-world example

A forecast of “2% GDP growth next year” is far less useful than “2%, with a 95% interval from −0.5% to 4.5%”. A bank planning its capital, or a government planning its budget, needs to know the plausible bad case, not just the middle. That is why the course keeps asking for uncertainty: standard errors, prediction intervals, Value-at-Risk.

1.4 What economic time series typically look like

The lecture lists a few patterns that keep showing up in economic and financial data. You will meet each of them again later in the book.

1. **Trends.** Many series grow (or shrink) smoothly over time, like GDP or population.
2. **Meandering (random-walk) behaviour.** Stock prices and exchange rates wander around without any fixed level to return to. Shocks to them seem to last for a very long time. We say they contain a *stochastic trend* (Chapter 5).
3. **Changing volatility.** Some series are calm for a while and then wild for a while. S&P 500 returns are very volatile in some stretches and tranquil in others; US GDP growth became much calmer after 1985 (the “Great Moderation”). Such series are called *conditionally heteroskedastic* (Chapter 8).
4. **Strong correlation between series.** Gold and silver prices move closely together (correlation 0.85). With many such series, we need special tools (Chapters 7 and 9).
5. **Seasonality.** Some months, weeks or days behave differently: gas consumption is higher in winter; stock returns on Mondays tend to be lower than on the preceding Friday (the “weekend effect”). Data are often *seasonally adjusted* before analysis.

1.5 Prices versus returns

For financial data there is one more habit to learn straight away. We almost never model *prices*; we model *returns*. Look at Figure 1.1: Tesla’s price wanders from \$25 to over \$900 — there is no “typical level”. Its daily returns, on the other hand, bounce around zero in a stable band.

We compute returns as **log returns**:

$$r_t = 100 \times (\ln P_t - \ln P_{t-1}).$$

In words: the percentage change from yesterday to today (for small changes, a log return is almost identical to the ordinary percentage change).

Log returns have a convenient property: they *add up* over time. The return over a week is simply the sum of the five daily returns. That is why the course always uses them.

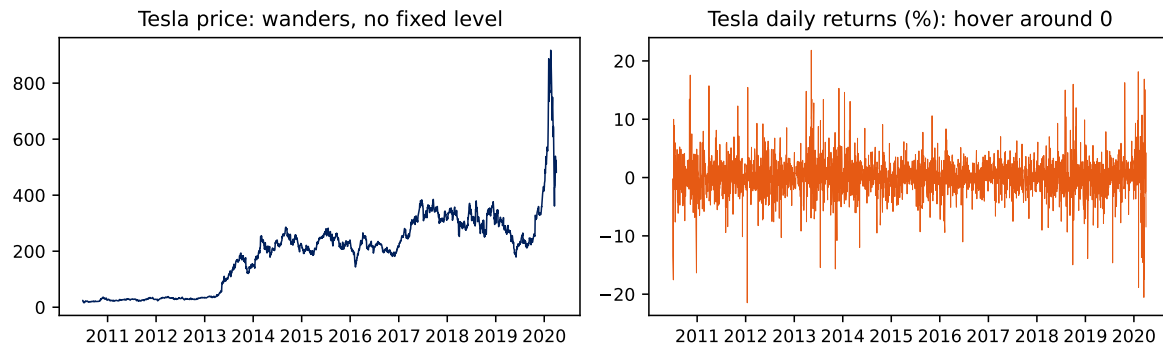


Figure 1.1: The same stock, two views. Prices wander; returns hover around a fixed level.

1.6 Describing a series: descriptive statistics

Before modelling, we summarise a series with a few numbers:

- **mean** and **median**: the typical value (the median is less affected by a few extreme observations);
- **standard deviation**: the typical size of the fluctuations (for returns: the *volatility*);
- **skewness**: asymmetry. Positive skewness means a long tail of large positive values; negative skewness, a long tail of large losses. A normal distribution has skewness 0;
- **kurtosis**: how heavy the tails are — how often extreme values occur. A normal distribution has kurtosis 3; above 3 means “fat tails”.

The **Jarque–Bera test** combines skewness S and kurtosis K into one test of normality:

$$JB = \frac{T}{6} \left(S^2 + \frac{(K - 3)^2}{4} \right), \quad H_0 : \text{the data are normally distributed.}$$

In words: JB is large when skewness is far from 0 or kurtosis far from 3. A large JB (small p -value) rejects normality.

Real-world example

The lecture shows monthly Google searches for “financial crisis” (2004–2014): skewness 3.7, kurtosis 21.3, Jarque–Bera 1,979. Most months nobody searches the term, then in late 2008 searches explode — a long right tail. Normality is overwhelmingly rejected. Tesla’s daily returns tell the same story in finance: kurtosis 9.5, Jarque–Bera 4,261. Days of -21% and $+22\%$ occur, which a normal distribution with Tesla’s volatility would essentially never produce.

1.7 The stylized facts of returns

When you study returns of almost any asset, the same handful of facts appear again and again. They are called **stylized facts**, and every model in this course exists to explain one of them:

1. **Returns are hard to predict from past returns** (their autocorrelation is close to zero). This is what the *efficient market hypothesis* predicts: if past returns predicted future returns, traders would exploit it until it disappeared.
2. **Fat tails.** Extreme days happen far more often than a normal distribution would suggest. (Tesla's kurtosis is 9.5; a normal distribution has 3.)
3. **Volatility clustering.** Big moves follow big moves, small moves follow small moves — in either direction.
4. **Leverage effect.** Bad news tends to raise volatility more than good news of the same size.

The idea in plain words

The whole course in one sentence: **the direction of returns has (almost) no memory, but the size of returns does.** You cannot predict whether tomorrow goes up or down, but you can predict whether tomorrow will be a calm or a wild day.

Chapter 2

Stationarity: The Most Important Word in the Course

2.1 The window test

Imagine you cut two short windows out of a time series — one from 2012 and one from 2019 — and remove the dates. If nobody could tell which window came from which year, because both look the same (same average level, same spread, same kind of wiggles), the series is **stationary**.

The idea in plain words

A series is **stationary** if its behaviour does not depend on *when* you look at it. Its average, its spread and its memory are the same in every period.

Tesla's price fails the window test instantly (around \$30 in 2012, around \$300 in 2019). Tesla's returns pass it: in both windows they hover around zero with a similar spread.

Real-world example

Think of the temperature in a room with a thermostat set to 21°C. It goes up when the sun shines in and down when a window is opened, but it is always pulled back towards 21°C. Its average and its typical spread are the same in January and in July: **stationary**. Now think of a hiker lost in fog, taking random steps. Where she is now is the sum of all her past steps; nothing pulls her back to where she started, and the longer she walks, the further away she may be. That is a **random walk** — non-stationary. Returns behave like the room; prices behave like the hiker.

2.2 The formal definition

The slides give two versions. **Strict stationarity** says the whole probability distribution of the series stays the same when you shift time. That is hard to check, so in practice we use the version based on the first two moments (often called *weak* or *covariance* stationarity):

1. the mean is constant: $E(y_t) = \mu$ for every t ;
2. the variance is constant: $\text{Var}(y_t) = \sigma^2$ for every t ;
3. the covariance between y_t and y_{t-k} depends only on the distance k , not on the date.

In words: same average, same spread, same memory — at all times.

Notice what is *allowed*: a stationary series can certainly have memory (today can depend on yesterday). What is not allowed is that this dependence changes over calendar time.

2.3 Strict versus covariance stationarity

When does the weaker, moment-based version imply the strict one? When the series is **normally distributed**. A normal distribution is completely described by its mean, variance and covariances, so if those are constant over time, the whole distribution is constant over time.

Covariance stationarity also requires the variance to *exist* (be finite). For a normal distribution all moments exist. For a Student- t distribution with ν degrees of freedom, the variance exists only if $\nu > 2$ (and the kurtosis only if $\nu > 4$). With very fat tails, “the variance” may not even be a finite number — one more reason why fat tails matter in finance.

2.4 Trends and stationarity

A **trend** means the typical level of the series changes over time. The lecture’s example is $y_t = 0.5 + t + \varepsilon_t$: the mean $0.5 + t$ grows every period, so condition 1 fails. A series with a trend is therefore *not* stationary.

Stationarity rules out more than trends, though. A series whose variance keeps changing (for example, it becomes much calmer after a certain year) is also non-stationary, even if its mean is constant. And a series with strong seasonality is non-stationary too, because December behaves differently from July — the date matters. When a series has both a trend and seasonality, both must be dealt with before using methods that require stationarity.

2.5 Why we care so much

Almost every tool in this course — ARMA, VAR, PCA, and the formulas for long-run means and variances — assumes the data are stationary. The reason is simple. When you compute “the average” of a series, you are assuming there *is* one average to estimate. If the series trends, its average is different every year, and the number you compute describes nothing.

The lecture shows a dramatic example: monthly Google searches for “financial crisis” and for “Shogun lyrics” move together almost perfectly between 2009 and 2014. Of course, the two have nothing to do with each other. Both simply happened to trend in the same period. **When unrelated series share strong trends, a regression can look meaningful even when it is not.** This is called a *spurious regression*, and we come back to it in Chapter 5.

Watch out

“Stationary” does not mean “boring” or “no dependence”. A stationary series can have strong memory. It just has to have the *same* memory, level and spread throughout.

Real-world example

Seasonality can fool a regression too. The lecture shows weekly Google searches for “econometrics” and for “art history”. They move together beautifully — not because econometricians love art, but because both follow the *academic calendar*: high during semesters, low in summer and around Christmas. Remove the seasonal pattern and the relationship vanishes.

2.6 What time dependence does to OLS

Your regression course gave you two sets of assumptions for OLS in $y_t = x_t'\beta + \varepsilon_t$:

- **Finite-sample (Gauss–Markov + normality):** errors have mean zero, are *completely independent of all the regressors* (A2), have constant variance and are uncorrelated, and are normal. Then OLS is *unbiased* and exactly normally distributed.
- **Asymptotic (large-sample):** weaker conditions — no normality needed, and the error only has to be independent of the *same-period* regressors (A8), plus a stable average of $x_t x_t'$. Then OLS is *consistent* and approximately normal, so the usual t - and F -tests still work in large samples.

The slide's question: if you explain today's inflation by yesterday's inflation, which assumption can hold? The regressor y_{t-1} was itself built from earlier shocks, so it cannot be independent of *all* errors (A2 fails). But today's shock can be unrelated to yesterday's inflation (A8 holds). **So OLS on an AR model is biased in small samples, but consistent** — fine with enough data.

Two other violations matter too. If the errors are *autocorrelated* or *heteroskedastic*, OLS stays consistent, but its standard errors are wrong, so t -tests and p -values mislead. A quick fix is **HAC standard errors** (heteroskedasticity- and autocorrelation-consistent). A better fix is to *model* the dependence directly: ARMA models for autocorrelation (Chapter 4), volatility models for heteroskedasticity (Chapter 8).

Real-world example: the first tutorial's simulation

In Tutorial 1 you simulated $y_i = 10 + 0.2x_i + \varepsilon_i$ with a variance that jumps from 1 to 4 halfway through the sample. One estimate came out at 0.126 — not 0.2. Repeating the experiment 1,000 times, the estimates averaged **0.203**: OLS is still *unbiased*. Increasing the sample from 30 to 3,200 observations shrank their spread from 0.30 to 0.03: still *consistent*. What heteroskedasticity damages is the standard errors and the efficiency, not the centre of the estimates. Simulation is the right tool here because only with simulated data do we *know* the true value.

Measuring Memory: Autocorrelation, ACF and PACF

3.1 Autocorrelation

Correlation measures how two variables move together. **Autocorrelation** applies the same idea to a single series and a delayed copy of itself. Take the series, shift it back by one period, and compute the correlation between the two: that is the autocorrelation at lag 1. Shift by k periods and you get the autocorrelation at lag k :

$$\rho_k = \text{Corr}(y_t, y_{t-k}) = \frac{\gamma_k}{\gamma_0},$$

where $\gamma_k = \text{Cov}(y_t, y_{t-k})$ is called the k -th *autocovariance* and γ_0 is simply the variance.

In words: ρ_k tells you how strongly today's value is linked to the value k periods ago.

- ρ_1 close to +1: high values follow high values — the series moves in long, smooth swings.
- ρ_1 negative: the series tends to zigzag.
- all ρ_k equal to zero: the past says nothing about the present. Such a series is called **white noise**.

3.2 The ACF and the PACF

The **ACF** (autocorrelation function) is simply the list $\rho_1, \rho_2, \rho_3, \dots$, drawn as a bar chart. It is a picture of the series' memory.

The **PACF** (partial autocorrelation function) answers a slightly sharper question. Today may be linked to two days ago only *because* today is linked to yesterday and yesterday is linked to the day before — the link is passed along a chain. The PACF at lag k measures the *direct* link between y_t and y_{t-k} , after removing what is passed through the lags in between. The slides compute it with a regression: to get PACF(2), regress y_t on y_{t-1} and y_{t-2} ; the coefficient on y_{t-2} is PACF(2).

The idea in plain words

ACF = total link with the past (including links passed along a chain). PACF = direct link only. Together they form a fingerprint that tells us which model fits the data (Chapter 4).

3.3 Is a bar big enough to matter?

Even a series with no memory at all will show small autocorrelations by pure chance. So every ACF plot comes with two dashed lines at

$$\pm \frac{1.96}{\sqrt{T}}.$$

Bars inside the dashed lines are indistinguishable from zero; only bars sticking out deserve attention. With $T = 2,456$ Tesla returns the band is ± 0.040 , and indeed all of Tesla's return autocorrelations fall inside it: no memory in returns. The autocorrelations of Tesla's *squared* returns, however, start at 0.185 — far outside the band. Same data, two very different answers: the direction has no memory, the size does.

Watch out

“Uncorrelated” is not the same as “independent”. A flat ACF only says there is no *linear* link with the past. Tesla returns are uncorrelated, yet clearly not independent, because their sizes are linked. Likewise, observations from an AR model share the same distribution, but they are *not* independent — each depends on the previous one.

3.4 Testing many lags at once: Ljung–Box

Looking at bars one by one invites cherry-picking. The **Ljung–Box test** checks the first m autocorrelations together:

$$Q = T(T+2) \sum_{k=1}^m \frac{\hat{\rho}_k^2}{T-k}, \quad H_0 : \text{no autocorrelation up to lag } m.$$

A large Q (small p -value) means there is memory left. Its most important use is on a model's *residuals*: a good model leaves white-noise residuals, so the test should *not* reject.

3.5 Reading ACFs from the course data

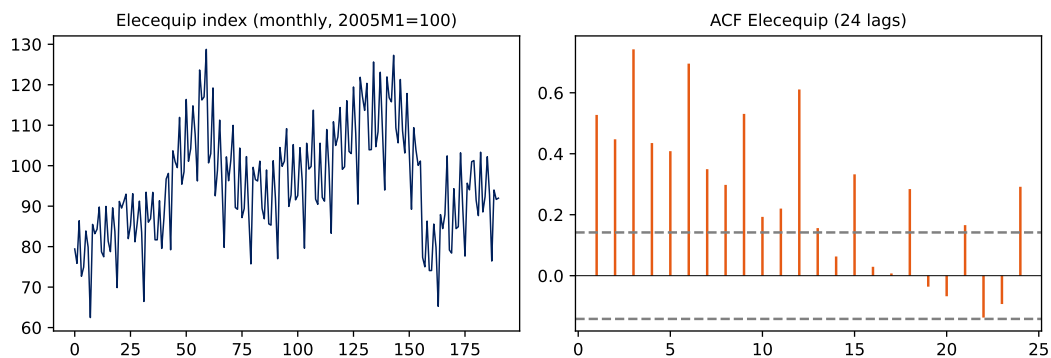


Figure 3.1: Euro-area electrical equipment production (monthly) and its ACF.

Real-world example: three ACFs from the tutorials

Electrical equipment production (Figure 3.1): autocorrelations of 0.53, 0.45 and 0.74 at lags 1–3, still 0.61 at lag 12, dying out very slowly. Slow decay warns of a possible stochastic trend; the bump at lag 12 is the *seasonal* pattern (this month resembles the same month last year).

Monthly pepper price changes: one clear bar (0.34 at lag 1) and nothing after — short, simple memory, typical of a stationary return series.

Tesla daily returns: every bar inside ± 0.040 — no memory. **Tesla squared returns:** 0.185, 0.176, 0.109, ... — strong, slowly fading memory in the *size* of moves.

Chapter 4

Models for One Series: AR, MA and ARMA

4.1 The autoregressive model, AR(1)

The simplest way to use the past is to say: *today equals a constant, plus a fraction of yesterday, plus something new*. That is the AR(1) model, written in the notation of the slides and the exam:

$$y_t = \alpha_0 + \alpha_1 y_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim NID(0, \sigma^2).$$

In words: today = constant + α_1 times yesterday + a fresh, unpredictable shock.

Two assumptions are always attached:

- the **exogeneity condition** $\text{cov}(y_{t-1}, \varepsilon_t) = 0$: today's shock is genuinely new, unrelated to the past;
- the **stationarity condition** $|\alpha_1| < 1$: yesterday's influence must fade. If $\alpha_1 = 1$, the influence never fades and we get a random walk (Chapter 5).

4.2 Deriving the long-run mean (a guaranteed exam skill)

The *long-run* or *unconditional* mean is the level the series hovers around. The slides show an “easy solution” that only needs three facts.

Long-run mean of an AR(1)

1. **Take expectations of both sides:** $E(y_t) = \alpha_0 + \alpha_1 E(y_{t-1}) + E(\varepsilon_t)$.
2. **Use white noise:** $E(\varepsilon_t) = 0$.
3. **Use stationarity:** the mean is the same every period, so $E(y_t) = E(y_{t-1}) = \mu$. This gives $\mu = \alpha_0 + \alpha_1 \mu$.
4. **Solve:** $\mu = E(y_t) = \frac{\alpha_0}{1 - \alpha_1}$

In words: the level the series settles around is the constant, scaled up by how much the series feeds on itself.

Example: $y_t = 2 + 0.6 y_{t-1} + \varepsilon_t$ has long-run mean $2/0.4 = 5$.

4.3 Deriving the long-run variance

Same idea, one extra trick.

Long-run variance of an AR(1)

1. Take the variance of both sides. The constant α_0 has no variance. Because $\text{cov}(y_{t-1}, \varepsilon_t) = 0$, the variance of the sum is the sum of the variances: $\text{Var}(y_t) = \alpha_1^2 \text{Var}(y_{t-1}) + \sigma^2$.

2. Use stationarity: $\text{Var}(y_t) = \text{Var}(y_{t-1}) = \gamma_0$, so $\gamma_0 = \alpha_1^2 \gamma_0 + \sigma^2$.

3. Solve:
$$\gamma_0 = \text{Var}(y_t) = \frac{\sigma^2}{1 - \alpha_1^2}$$

In words: the series is more variable than its shocks, because each shock keeps echoing for a while.

Notice: both formulas break down when $\alpha_1 = 1$ (division by zero). That is the mathematical face of non-stationarity. Note also *which assumptions you used*: white noise errors, exogeneity, and stationarity ($|\alpha_1| < 1$). Stating them is part of a full exam answer.

Real-world example: inflation as an AR(1)

Suppose monthly inflation (in %, annualised) follows $y_t = 0.2 + 0.9y_{t-1} + \varepsilon_t$, with $\sigma = 0.3$.

- **Long-run mean:** $0.2 / (1 - 0.9) = 2\%$ — conveniently, the ECB's target.
- **Long-run variance:** $0.09 / (1 - 0.81) = 0.47$, so the long-run standard deviation is about 0.69 points.
- **Forecasts after an inflation shock:** if inflation is 5% today, the forecasts are $0.2 + 0.9 \times 5 = 4.7\%$ next month, then 4.43%, 4.19%, ..., creeping back to 2%.
- **Half-life:** a deviation from 2% halves after $\ln 0.5 / \ln 0.9 \approx 6.6$ months.

A coefficient of 0.9 means high persistence: a shock is mostly still there next month. With $\alpha_1 = 0.3$, the same shock would be practically gone within three months.

4.4 The same derivation with an extra variable

The exam likes to add an explanatory variable x_t (which is stationary, with mean $E(x_t) = \mu_x$):

$$y_t = \alpha_0 + \alpha_1 y_{t-1} + \beta_1 x_t + \varepsilon_t.$$

Nothing new is needed — just carry the extra term along.

Mean. Taking expectations: $\mu = \alpha_0 + \alpha_1 \mu + \beta_1 \mu_x$, so

$$E(y_t) = \frac{\alpha_0 + \beta_1 \mu_x}{1 - \alpha_1}.$$

Variance. Taking variances, and using the exogeneity conditions $\text{cov}(y_{t-1}, \varepsilon_t) = \text{cov}(x_t, \varepsilon_t) = 0$:

$$\text{Var}(y_t) = \alpha_1^2 \text{Var}(y_{t-1}) + \beta_1^2 \text{Var}(x_t) + \sigma^2 + 2\alpha_1 \beta_1 \text{cov}(y_{t-1}, x_t).$$

If we *assume* that x_t is unrelated to last period's y (so the last covariance is zero), stationarity

gives

$$\boxed{\text{Var}(y_t) = \frac{\beta_1^2 \sigma_x^2 + \sigma^2}{1 - \alpha_1^2}}, \quad \sigma_x^2 = \text{Var}(x_t).$$

Exam corner

The exam instruction says *clarify any assumptions*. In this derivation, write explicitly: “I assume $\text{cov}(y_{t-1}, x_t) = 0$ ”. That one sentence turns a doubtful answer into a full-mark answer.

4.5 A second route: the MA(∞) form

The slides also derive the mean the “long way”, which shows *why* $|\alpha_1| < 1$ is needed. Substitute the model into itself again and again:

$$y_t = \alpha_0 + \alpha_1(\alpha_0 + \alpha_1 y_{t-2} + \varepsilon_{t-1}) + \varepsilon_t = \cdots = \alpha_0 \sum_{k=0}^{\infty} \alpha_1^k + \sum_{k=0}^{\infty} \alpha_1^k \varepsilon_{t-k} + \lim_{k \rightarrow \infty} \alpha_1^k y_{t-k}.$$

In words: today is a constant, plus all past shocks with fading weights, plus a leftover from the distant past.

The leftover term vanishes, and the sum $\sum \alpha_1^k = 1/(1 - \alpha_1)$ is finite, *only if* $|\alpha_1| < 1$. Taking expectations then gives $E(y_t) = \alpha_0/(1 - \alpha_1)$ — the same answer as the short route. This representation (“today = a weighted sum of all past shocks”) is called the MA(∞) form of the AR(1).

4.6 How strong is the memory?

In an AR(1), the autocorrelation at lag k is $\rho_k = \alpha_1^k$. With $\alpha_1 = 0.6$: 0.6, 0.36, 0.22, ... — the memory fades geometrically. That is why the ACF of an AR(1) *decays slowly* rather than stopping suddenly.

4.7 Moving average models, MA(q)

A moving average model builds today from *shocks* rather than past values:

$$y_t = \alpha_0 + \varepsilon_t + \beta_1 \varepsilon_{t-1} + \cdots + \beta_q \varepsilon_{t-q}.$$

In words: today = a constant + today’s news + echoes of the last q pieces of news.

For the MA(1), $y_t = \alpha_0 + \varepsilon_t + \beta_1 \varepsilon_{t-1}$, the moments are easy: the mean is α_0 (all shocks have mean zero) and, because shocks are uncorrelated, the variance is $(1 + \beta_1^2)\sigma^2$. Two observations more than one period apart share no shock at all, so their correlation is exactly zero. That gives the MA’s fingerprint: its ACF **cuts off** after lag q . An MA model is always stationary, because it is a finite sum of stationary shocks.

Why would past shocks matter? The slides admit the MA model is less intuitive than the AR model. Three realistic stories:

Real-world example

Overreaction and correction. A surprising earnings announcement moves a stock 5% today; part of the move is reversed tomorrow as traders digest the news. Today's shock leaves a (negative) echo one day later: an MA(1) with $\beta_1 < 0$.

Bid–ask bounce. In very high-frequency data, trades alternate between the bid and the ask price, so a recorded “up” move tends to be followed by a “down” move — again a negative MA(1) pattern.

A one-off disruption. A strike closes a factory for one month: output drops this month, and next month there is a catch-up. The effect of the shock lasts exactly one extra period, then is gone.

MA models also have an **estimation issue**: the past errors $\varepsilon_{t-1}, \varepsilon_{t-2}, \dots$ are not observed, so we cannot simply run OLS. That is why MA and ARMA models are estimated by maximum likelihood.

4.8 ARMA, and the family of abbreviations

Combine the two and you get an ARMA(p, q) model: p lags of y and q lags of the shock. The slides list the family:

AR(p)	autoregressive with p lags
MA(q)	moving average with q lags
ARMA(p, q)	both together
ARMAX(p, q)	ARMA plus extra explanatory variables X
ARIMA(p, d, q)	ARMA for data that must be differenced d times (non-stationary)

The course focuses on the first four.

4.9 The general AR(p) model

With p lags, $y_t = \alpha_0 + \alpha_1 y_{t-1} + \dots + \alpha_p y_{t-p} + \varepsilon_t$, where the errors are white noise: $E(\varepsilon_t) = 0$, $E(\varepsilon_t^2) = \sigma^2$, and $E(\varepsilon_t \varepsilon_s) = 0$ for $t \neq s$. It is stationary if the roots of its “characteristic polynomial” lie outside the unit circle — you do not need to compute these; the course only asks you to recognise the AR(1) case $|\alpha_1| < 1$, and to see that $\alpha_1 = 1$ (with $\alpha_0 = 0$) is the random walk $y_t = y_{t-1} + \varepsilon_t$. An AR(1) is the same as an ARMA(1,0).

4.10 Choosing the model

Method 1: read the ACF and PACF. Each model leaves a fingerprint (Figure 4.1):

Model	ACF	PACF
AR(p)	decays slowly	cuts off after lag p
MA(q)	cuts off after lag q	decays slowly
ARMA(p, q)	decays	decays

Method 2: information criteria. A bigger model always fits the sample better, so we need a

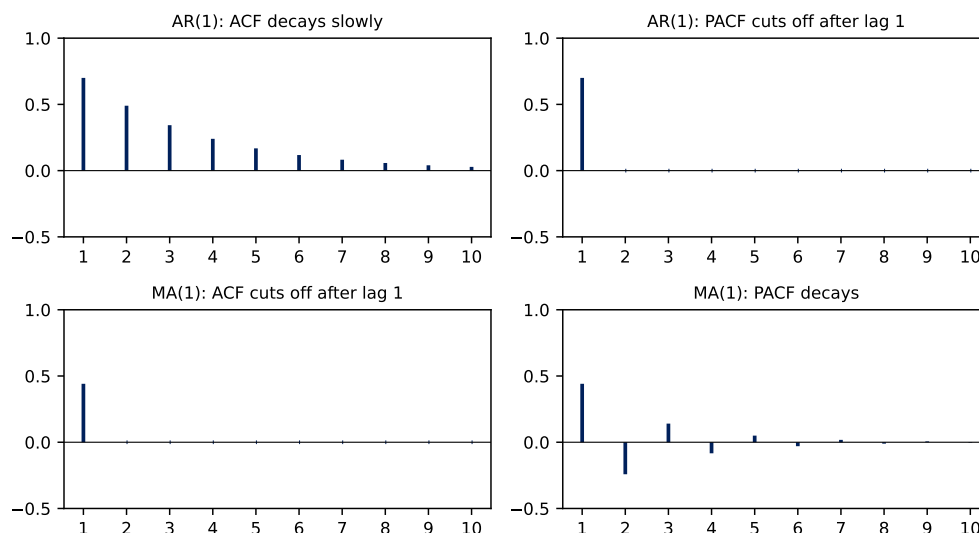


Figure 4.1: The fingerprints: AR shows its order in the PACF, MA shows its order in the ACF.

score that rewards fit but charges a “rent” for every parameter:

$$AIC = -2 \ln L + 2k, \quad BIC = -2 \ln L + k \ln T,$$

where L is the likelihood (how well the model fits) and k the number of parameters. **Smaller is better.** BIC charges more rent per parameter (as soon as $T > 8$), so it prefers smaller models than AIC.

Estimation. AR models can be estimated by OLS. MA and ARMA models are estimated by *maximum likelihood* (ML): the computer searches for the parameter values that make the observed data most likely. The slides list ML’s advantage (it works for almost any model) and its disadvantages (results depend on the error assumptions, the search can depend on starting values, and complex models can be hard to estimate).

Real-world example: when the “best” model loses

In the ARMA tutorial, four models were fitted to squared Nasdaq returns. Both AIC and BIC ranked the biggest model, an ARMA(5,5) with eleven parameters, as the best. Then the last 150 days were held back and forecast. The ARMA(5,5) scored an MSFE of 5.74 — *worse* than simply forecasting the historical average every day (5.68). In the Tesla tutorial, the same ARMA(5,5) did not even *converge*: the estimation algorithm gave up, so its numbers were meaningless. Two lessons: always check that estimation converged, and let out-of-sample performance have the final word. A big model can fit the past by memorising noise.

4.11 When the model is wrong

A good model leaves residuals that look like white noise — nothing left to explain. If the residuals are still autocorrelated (check their ACF, or the Ljung–Box test), the model is *misspecified*. The slides give two cures: (1) increase p or q ; (2) add explanatory variables (an ARMAX model).

4.12 Forecasting

A forecast is our best guess of the future given everything we know today (the information set I_t). For the AR(1):

$$E(y_{t+1} | I_t) = \alpha_0 + \alpha_1 y_t, \quad E(y_{t+2} | I_t) = \alpha_0 + \alpha_1 E(y_{t+1} | I_t) = \alpha_0 + \alpha_0 \alpha_1 + \alpha_1^2 y_t.$$

In words: forecast tomorrow from today; then forecast the day after from your forecast of tomorrow.

The forecast errors grow with the horizon: the one-step error is ε_{t+1} (variance σ^2), the two-step error is $\varepsilon_{t+2} + \alpha_1 \varepsilon_{t+1}$ (variance $\sigma^2(1 + \alpha_1^2)$), and so on. As the horizon grows, the forecast settles at the long-run mean and the error variance settles at the long-run variance $\sigma^2 / (1 - \alpha_1^2)$. In plain words: *far into the future, the best a stationary model can say is “the average, give or take the usual spread”*.

To compare forecasts we use the **MSFE** (mean squared forecast error) and the **MAFE** (mean absolute forecast error) over F forecast periods:

$$MSFE = \frac{1}{F} \sum_{f=1}^F (y_f - \hat{y}_f)^2, \quad MAFE = \frac{1}{F} \sum_{f=1}^F |y_f - \hat{y}_f|.$$

Lower is better.

4.13 Forecast horizons and forecasting in practice

How far ahead to forecast depends on the decision: a daily trader needs 1-day-ahead forecasts; a pension fund cares about months or years. **Accuracy typically falls as the horizon grows**, because errors accumulate and the forecast slides towards the uninformative long-run mean.

Two ways to produce forecasts over a test period:

- **Fixed origin:** estimate once, then forecast $1, 2, \dots, H$ steps ahead from that single point.
- **Rolling or expanding origin:** move forward one period at a time, re-estimate (on a rolling window or on all data so far) and forecast one step ahead each time. These forecasts use only information available at the time — honest “real-time” forecasts.

Real-world example: one MSFE number can hide the story

In the R tutorial, four models forecast Tesla’s squared returns for 150 days from 27 August 2019. The plot of *cumulative* squared errors stayed low and flat for 100 days — then shot up: the COVID crash of March 2020 produced about 90% of all forecast error. Before COVID, a different model was winning. A single MSFE number would hide both facts. Lesson: a forecast evaluation should show *when* errors happen, not only how big they are on average.

Chapter 5

Trends and Unit Roots

5.1 Two very different kinds of trend

A series can drift upward for two completely different reasons. Figure 5.1 builds three series from *exactly the same random shocks*.

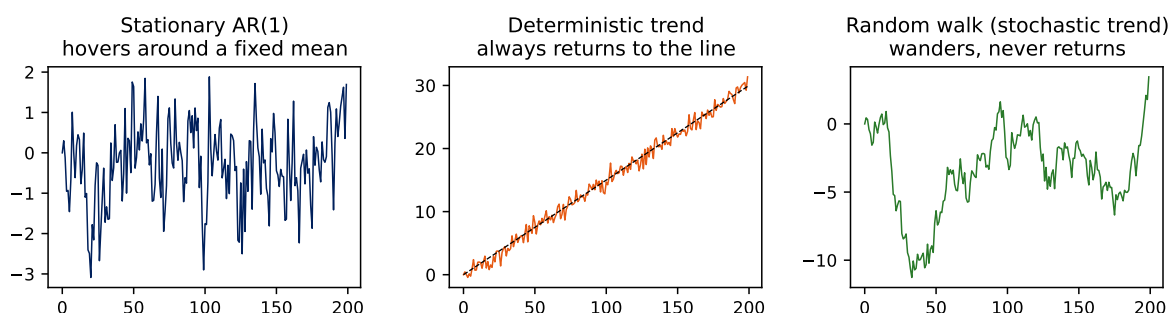


Figure 5.1: Same shocks, three behaviours.

Deterministic trend: $y_t = \alpha + \beta t + \varepsilon_t$. The series wobbles around a straight line. Shocks push it off the line for one period only; it always comes back.

Stochastic trend (random walk): $y_t = y_{t-1} + \varepsilon_t$ (possibly plus a drift). Each period adds a new shock *on top of* everything before. The lecture's image is *the drunkard's walk*: a man walking home after a night out takes random steps left and right; where he ends up is the sum of all his random steps, and he never "returns" anywhere in particular. Stock prices behave like this.

5.2 Telling trends apart is hard

The lecture opens with a quiz: twelve series, which ones have trends? The lesson is humbling. *Different* mechanisms (a straight line plus noise, or a random walk) can produce trends that look alike, and the *same* mechanism (a random walk) can produce series that look completely different from one run to the next.

Random walks can fool experts. A "stupidly simple" random walk can produce curves that look complicated — bends, booms, apparent regime changes. The lecture's warning: a machine-learning expert may see such a curve and say "this is clearly nonlinear, we need neural networks!", when it is just a random walk.

Stochastic trends are everywhere. Stock prices behave like random walks, and so (roughly) do many economic series: GDP, unemployment, consumption, exchange rates. A random walk is unpredictable in a precise sense: the most likely next value is simply today's value.

Both trends at once. If the random walk's steps have a non-zero average μ (a *drift*), write

$$u_t = \mu + \varepsilon_t:$$

$$x_t = x_0 + (\mu + \varepsilon_1) + \cdots + (\mu + \varepsilon_t) = x_0 + \underbrace{\mu t}_{\text{deterministic trend}} + \underbrace{\varepsilon_1 + \cdots + \varepsilon_t}_{\text{stochastic trend}}.$$

In words: a random walk with drift contains both kinds of trend: a straight-line climb from the drift, and wandering from the accumulated shocks.

Stock prices are a good example: a positive average return (drift) plus random shocks.

Two complicating factors (slides): a *nonlinear* deterministic trend can look like a stochastic trend, and a stationary but *highly persistent* series (say $\alpha_1 = 0.98$) looks almost exactly like a random walk in a finite sample.

5.3 What happens to a shock?

This is the cleanest way to tell the two apart. Take the AR model $x_t = \rho x_{t-1} + \varepsilon_t$ and substitute it into itself again and again:

$$x_t = \varepsilon_t + \rho \varepsilon_{t-1} + \rho^2 \varepsilon_{t-2} + \cdots + \rho^{t-1} \varepsilon_1 + \rho^t x_0.$$

In words: a shock that happened k periods ago still affects today with weight ρ^k .

- If $|\rho| < 1$: the weight ρ^k shrinks towards zero. Shocks **die out**; the series returns to its mean. This is called **mean reversion**, and it is what makes a series stationary.
- If $\rho = 1$ (random walk): the weight is $1^k = 1$ forever. Shocks are **permanent**. There is no mean to return to.

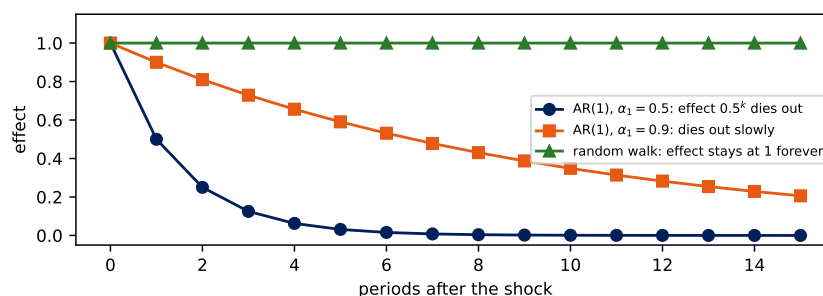


Figure 5.2: The effect of one shock over time.

Two consequences of $\rho = 1$ to remember: the variance of a random walk grows with time ($\text{Var}(y_t) = t\sigma^2$), and its best forecast is simply today's value, with uncertainty that keeps growing the further ahead you look.

A process with $\rho = 1$ is called a **unit root** process, or **integrated of order one**, written $I(1)$. A stationary series is $I(0)$. A series that must be differenced twice to become stationary is $I(2)$.

5.4 Removing a trend

Each kind of trend has its own cure.

- **Deterministic trend:** regress the series on a time trend and keep the residuals, $\hat{u}_t = x_t - \hat{\alpha} - \hat{\beta}t$ (“detrending”).
- **Stochastic trend:** take the **first difference**, $\Delta x_t = x_t - x_{t-1}$. For a random walk, $\Delta x_t = \varepsilon_t$ — pure, stationary noise. Detrending alone does *not* remove a stochastic trend.

This is exactly why we turn prices (I(1)) into returns (I(0)): a log return is the first difference of log prices.

Forecasting with a stochastic trend (slide recipe): difference the series; forecast the differences, e.g. with an ARMA model including an intercept; then add the forecast changes back onto the last observed level.

Forecasting with a linear trend (slide recipe):

1. remove the trend: $\hat{u}_t = x_t - \hat{\alpha} - \hat{\beta}t$;
2. forecast the detrended series $\hat{u}_{n+1}, \dots, \hat{u}_{n+h}$, e.g. with an ARMA model;
3. add the trend back: $\hat{x}_{n+h} = \hat{u}_{n+h} + \hat{\alpha} + \hat{\beta}(n+h)$.

Notice how differently the two trend types forecast: a deterministic trend says “return to the line”, a stochastic trend says “start from where we are now”. Picking the wrong type gives completely different forecasts.

5.5 The danger of ignoring trends: spurious regression

Regress one random walk on another, completely unrelated random walk, and you will very often find a “significant” coefficient and a high R^2 . Both series simply wander, and over a finite sample they happen to wander in similar directions. The usual t -tests assume stationary data, so they are badly misleading here. This is the **spurious regression** problem — the “financial crisis” versus “Shogun lyrics” example of Chapter 2. The lesson: *check for unit roots before you regress one level on another*.

A useful rule for combinations: adding a stationary series to an I(1) series gives an I(1) series (“I(1) + I(0) = I(1)”). Stationary noise cannot cancel a stochastic trend. So if y_t depends on a non-stationary x_t (with a non-zero coefficient), then y_t is non-stationary too.

Real-world examples: spurious relations

Cheese and bedsheets. Between 2000 and 2009, US per-capita cheese consumption and the number of people who died by becoming tangled in their bedsheets rose together. A regression of one on the other gives a t -statistic of 19 and an R^2 of 0.88. Both simply trended upward over the same decade.

Nowcasting Dutch unemployment. The lecture tries to predict Dutch unemployment (CBS data) from Google searches for words like “CV”, “motivatiebrief” and “uitkering”

(benefits), using boosting, one of the best machine-learning methods. In-sample the fit is excellent; out of sample the forecasts are terrible. Re-estimating with data up to 2015 does not help, and neither does switching to neural networks or other methods. The culprit is not the method but the **trends**: the search series and unemployment trended, so the model learned a spurious relationship.

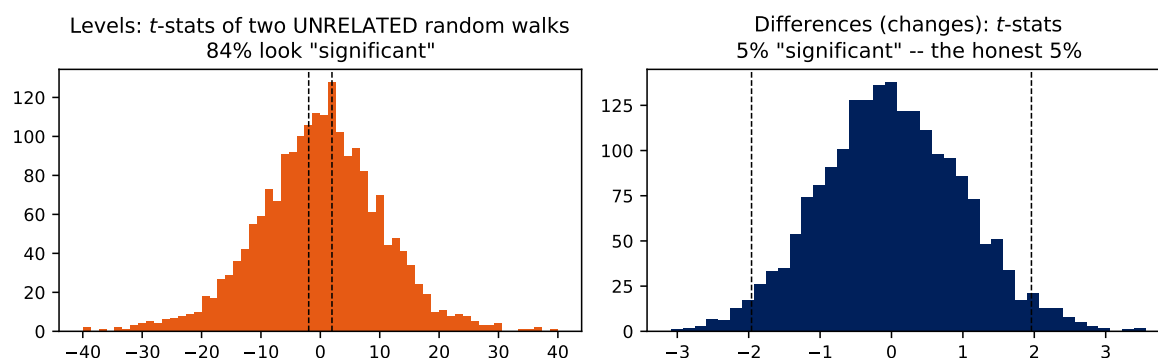


Figure 5.3: 2,000 regressions of one random walk on another, completely unrelated one. In levels, about 84% of the t -statistics are “significant”. After differencing, only the honest 5% are.

Figure 5.3 makes the danger concrete: with trending data, a regression finds a “significant” relationship most of the time even when none exists. Differencing (working with changes, like returns) restores honest inference.

5.6 Testing for a unit root: the Dickey–Fuller test

We want to test whether $\rho = 1$ in $x_t = \rho x_{t-1} + u_t$. Subtract x_{t-1} from both sides, and add a constant and a trend:

$$\Delta x_t = \alpha + \beta t + \delta x_{t-1} + \varepsilon_t, \quad \delta = \rho - 1.$$

$$H_0 : \delta = 0 \text{ (unit root, stochastic trend)} \quad H_1 : \delta < 0 \text{ (stationary)}$$

In words: if the series is far above its level and $\delta < 0$, it tends to come back down — that pull-back is mean reversion. If $\delta = 0$, there is no pull-back at all.

Two practical points from the slides:

1. The test statistic does *not* follow the usual t -distribution under H_0 . You must use the special **Dickey–Fuller critical values** (roughly -2.86 with a constant, -3.41 with constant and trend, at 5%). Reject H_0 only if the statistic is *more negative* than the critical value.
2. If the series may have a linear trend, include it — otherwise the test mistakes a deterministic trend for a stochastic one. The slides’ advice: *it is always safe to add the constant and the trend*.

The four possible outcomes (slide table). With both constant and trend included, the test regression tells you about both kinds of trend at once:

Result in $\Delta x_t = \alpha + \beta t + \delta x_{t-1} + \varepsilon_t$	Deterministic trend?	Stochastic trend?
$\beta = 0, \delta < 0$	no	no
$\beta = 0, \delta = 0$	no	yes
$\beta \neq 0, \delta < 0$	yes	no
$\beta \neq 0, \delta = 0$	yes	yes

Augmented Dickey–Fuller (ADF). If the errors are autocorrelated, add lagged differences: $\Delta x_t = \alpha + \beta t + \delta x_{t-1} + \gamma_1 \Delta x_{t-1} + \dots + \gamma_p \Delta x_{t-p} + \varepsilon_t$, with p chosen by AIC or BIC. The γ 's are not interpreted; they only fix the test.

KPSS is a test with the hypotheses the other way around: H_0 is *stationarity*. ADF and KPSS can disagree.

The lecturer's rules of thumb: test every variable for a unit root; difference the ones with a unit root; include a trend whenever you even slightly suspect trending behaviour; and *if in doubt, take differences*.

Watch out

The most common mistake on this topic is writing the hypotheses backwards. In Dickey–Fuller, the **unit root is the null hypothesis**. Failing to reject means “no evidence against a unit root” — not “proof” of one.

Rule 5, for large datasets: manual analysis is impossible with hundreds of series, so loop over all of them and decide with the ADF test with trend. Some series will be misclassified — that is inevitable. Use a small significance level so that you rarely reject a unit root by mistake.

Real-world example: the test depends on the specification

In the unit-root tutorial, the electrical-equipment series (Figure 3.1) gave two different answers. With a constant only, the ADF statistic was -2.92 against a 5% critical value of -2.86 : a unit root is (just) rejected. With constant *and* trend, it was -2.80 against -3.41 : not rejected. The series rises *and* falls, so it has no clear linear trend; including one makes the test stricter. Honest conclusion: weak evidence of stationarity, and the verdict depends on the specification — following the lecturer's rule “if in doubt, difference”, a cautious analyst would difference it. By contrast, Tesla's log *price* clearly fails to reject a unit root, while its *returns* reject it overwhelmingly: prices are $I(1)$, returns are $I(0)$.

5.7 Cointegration in one paragraph

Sometimes two $I(1)$ series wander, but wander *together*, so that some combination of them is stationary (for example, the spot and futures price of the same commodity). They are then called **cointegrated**, and the relationship between their levels is real, not spurious. Cointegration only makes sense for series that *have* stochastic trends. Series that are stationary around a deterministic trend are not candidates for cointegration — for them you simply include the trend.

5.8 The efficient market hypothesis and unit roots

If log prices follow a random walk, $\ln p_t = \beta_0 + \ln p_{t-1} + \varepsilon_t$, then returns are $r_t = \beta_0 + \varepsilon_t$: a constant plus unpredictable noise. Nothing from the past helps predict today's return — exactly what the efficient market hypothesis says. So in the regression $\ln p_t = \beta_0 + \beta_1 \ln p_{t-1} + \varepsilon_t$, the estimate supporting market efficiency is $\beta_1 = 1$. In a regression of *returns* on lagged returns, efficiency predicts a coefficient of *zero*.

Chapter 6

Extending a Model: The Exam's Favourite Move

Many exam questions say: “We suspect effect X . Extend the model to allow for it. Clearly state which parameter reflects the effect.” This chapter gives you one recipe that answers all of them.

The four-step recipe

1. **Build a variable that switches the effect on.** Usually a *dummy*: $D_t = 1$ when the special condition holds, 0 otherwise.
2. **Put it where the effect lives.** An effect on the *level* of returns goes into the mean equation. An effect on the *volatility* goes into the variance equation.
3. **Name the parameter and its expected sign.** “The parameter γ measures the effect; we expect $\gamma < 0$.”
4. **Say how you would test it.** “A t -test of $H_0 : \gamma = 0$.”

6.1 An effect on the level (the mean)

Lunch effect on returns (sample exam). Intraday 5-minute returns may be smaller during lunch. Define $D_t = 1$ if observation t falls between 12:00 and 13:00, and 0 otherwise. Add it to the mean equation:

$$y_t = \alpha_0 + \alpha_1 y_{t-1} + \beta_1 x_t + \beta_2 \text{trend}_t + \gamma D_t + \varepsilon_t.$$

γ is the lunch effect: the extra average return during lunch. Smaller lunch returns mean $\gamma < 0$.

The same recipe covers the **weekend effect** ($D_t = 1$ on Mondays), and the “**sell in May and go away**” effect ($D_t = 1$ for June–October; in a VAR, the dummy enters every equation with its own coefficient).

6.2 More calendar effects, and the dummy-variable trap

Researchers have found (and argued about) many calendar patterns, all handled with the same dummy recipe:

- the **January effect** (small-company stocks historically did well in January): $D_t = 1$ in January;
- **holiday effects** (returns before public holidays): $D_t = 1$ on the trading day before a holiday;
- **day-of-the-week effects**: one dummy per weekday.

Watch out

The dummy-variable trap. With five weekdays and a constant in the model, include only *four* weekday dummies (for example, Tuesday to Friday). If you include all five, they always add up to 1 — exactly the constant — and the model cannot be estimated. The left-out day (Monday) becomes the benchmark, and each dummy's coefficient measures the difference from Monday.

6.3 An effect on the volatility (the variance)

Lunch effect on volatility (sample exam). Now returns are suspected to be *calmer* at lunch, not lower. The effect must go into the *variance* of the error term. Write the error with a time-varying variance:

$$\varepsilon_t = \sigma_t v_t, \quad v_t \sim NID(0, 1), \quad \sigma_t^2 = \exp(\omega + \lambda D_t).$$

λ is the lunch effect on volatility; calmer lunches mean $\lambda < 0$. The exponential is a smart choice: whatever the values of ω and λ , the variance stays positive. (The simpler version $\sigma_t^2 = \omega + \lambda D_t$ also works, but then you must require $\omega > 0$ and $\omega + \lambda > 0$.)

Real-world example: a crisis dummy in the variance

Returns in February–April 2020 were far more volatile than usual (Tesla's daily volatility rose from about 3.4% to 8.5%). A dummy $C_t = 1$ during the COVID crash, placed in the variance equation, $\sigma_t^2 = \exp(\omega + \lambda C_t)$, would capture this with $\lambda > 0$. A dummy only captures a *known, dated* event, though. The volatility models of Chapter 8 go further: they let volatility rise and fall on its own, driven by recent shocks.

6.4 Volatility affecting the mean

Sometimes risk itself moves expected returns: investors demand a higher return in riskier periods. Put the variance into the mean equation ("ARCH-in-mean"):

$$y_t = \mu + \delta \sigma_{t-1}^2 + \eta_t.$$

δ is the effect of last period's volatility on today's mean return; a risk premium means $\delta > 0$. (Chapter 8 places this in the full volatility model.)

6.5 Is the trend really needed? Four ways to check

The sample exam asks for four methods to decide whether a deterministic trend belongs in a model. For each, say what outcome means "yes, keep it":

1. **t-test on the trend coefficient:** test $H_0 : \beta_2 = 0$. If rejected (e.g. $|t| > 1.96$, $p < 0.05$), the trend is needed.
2. **Information criteria:** estimate the model with and without the trend. If AIC/BIC are *lower* with the trend, keep it.

3. **Out-of-sample forecasting:** compare MSFE (or MAFE) of the two models on data not used for estimation. If the trend model forecasts better (lower MSFE), keep it.
4. **Look at the data, or use the Dickey–Fuller regression with trend:** a visible upward or downward drift in the plot, or a significant β in $\Delta x_t = \alpha + \beta t + \delta x_{t-1} + \varepsilon_t$, indicates a deterministic trend.

(An F -test, Wald test or likelihood-ratio test comparing the two models is a fine alternative to the t -test.)

6.6 What the efficient market hypothesis predicts

If the model is applied to stock *returns*, the efficient market hypothesis says past returns cannot predict future returns, so the coefficient on lagged returns should be $\alpha_1 = 0$. It also says returns should not drift systematically upward over time — a deterministic trend in returns would be a predictable, ever-growing profit — so $\beta_2 = 0$.

Exam corner

Phrase extension answers in three sentences: (1) “I define $D_t = 1$ if ..., 0 otherwise.” (2) “The extended model is ...” (write it out in full). (3) “The parameter ... reflects the effect; the hypothesis predicts it is negative/positive, which I can test with a t -test.”

Several Series Together: VAR Models

7.1 Why model series jointly?

So far each model described one series using its own past. But financial variables talk to each other: futures prices help predict spot prices; exchange rates in one market help predict another; inflation, output and money supply push each other around. A **Vector Autoregression (VAR)** lets every variable depend on the past of *every* variable.

The slides give two reasons to model series together:

1. **Exploiting dependence.** If series are interdependent, using them jointly improves estimates and forecasts — as if the combined data gave you more effective observations.
2. **Unobserved contemporaneous variables.** Suppose y_t depends on x_t , but x_t is not yet known when you must forecast y_t . Then x_t has to be modelled *together* with y_t .

Real-world example

Spot and futures. Futures prices react to news quickly; spot prices of the same commodity follow. Modelling both together lets yesterday's futures move help forecast today's spot price.

Currency markets. The euro-dollar rate in the Tokyo session can help predict its move in the London session; a VAR of the two exploits that.

Macro policy. Inflation, output and the policy rate all respond to each other's past — a central bank's forecasting model is typically a VAR of exactly this kind.

7.2 The two-variable VAR(1), written out

This is exactly the form used in the sample exam:

$$\begin{pmatrix} y_t \\ x_t \end{pmatrix} = \begin{pmatrix} \delta_1 \\ \delta_2 \end{pmatrix} + \begin{pmatrix} \theta_{11} & \theta_{12} \\ \theta_{21} & \theta_{22} \end{pmatrix} \begin{pmatrix} y_{t-1} \\ x_{t-1} \end{pmatrix} + \begin{pmatrix} \varepsilon_{1,t} \\ \varepsilon_{2,t} \end{pmatrix}.$$

Matrices can look intimidating, so multiply it out. It is just **two ordinary regressions**:

$$\begin{aligned} y_t &= \delta_1 + \theta_{11} y_{t-1} + \theta_{12} x_{t-1} + \varepsilon_{1,t}, \\ x_t &= \delta_2 + \theta_{21} y_{t-1} + \theta_{22} x_{t-1} + \varepsilon_{2,t}. \end{aligned}$$

In words: θ_{11} : how y depends on its own past. θ_{12} : how y depends on the past of x . θ_{21} : how x depends on the past of y . θ_{22} : how x depends on its own past.

Reading tip: the first index is the equation (who responds), the second is the variable (who causes). θ_{12} = “ y responding to x ”.

In compact vector notation, with $z_t = (y_t, x_t)'$: $z_t = \delta + \Theta z_{t-1} + \varepsilon_t$. Each equation has the same right-hand-side variables, so each can be estimated by OLS on its own.

7.3 The assumptions behind a VAR

All the AR assumptions carry over: the data are stationary, the errors are white noise and uncorrelated over time, and the usual exogeneity conditions hold (the lagged variables are uncorrelated with today's errors). One new ingredient: the two errors may be correlated *in the same period*. That is captured by the error covariance matrix

$$\Sigma = \begin{pmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{12} & \sigma_2^2 \end{pmatrix}, \quad \varepsilon_t \sim N(0, \Sigma),$$

where σ_{12} is the covariance between $\varepsilon_{1,t}$ and $\varepsilon_{2,t}$. Economically: news that hits one market today often hits the other market today too. When the sample exam says the errors are uncorrelated, it means $\sigma_{12} = 0$.

With p lags or more variables, the idea is identical: each variable is regressed on p lags of *all* variables. A three-variable VAR (say inflation, output and money) has three equations, each with three variables' lags on the right-hand side. Stationarity is checked with the "inverse roots", which must all lie inside the unit circle (R draws this circle for you).

7.4 Stationarity, restrictions and model choice

Stationarity works like in the AR(1), in matrix form: the influence of the past must fade. (Technically, the eigenvalues of Θ must lie inside the unit circle; the tutorial shows how to check it in R.)

Structural VAR. When economic theory says some links do not exist, we set those coefficients to zero instead of estimating them. Example from the slides: with inflation, output and money supply, one may argue that money affects inflation but not the other way around.

Choosing between VAR models (the slides' list):

1. start large and simplify using t -tests, F -tests or Wald tests;
2. compare R^2 — but only between models that are *not nested* (a bigger nested model always has a higher R^2);
3. compare AIC or BIC (works for nested and non-nested models);
4. check for misspecification: choose a model whose errors show no remaining autocorrelation;
5. compare forecast performance (MSFE, MAFE).

If the errors are still autocorrelated, the cures are: add lags, add variables to the system, or add exogenous variables (a VAR-X).

7.5 Forecasting

Exactly like the AR(1), but for all variables at once:

$$\hat{z}_{T+1} = \delta + \Theta z_T, \quad \hat{z}_{T+2} = \delta + \Theta \hat{z}_{T+1} = \delta + \Theta \delta + \Theta^2 z_T.$$

7.6 Impulse responses

The idea in plain words

An **impulse response** answers: “If one variable receives a one-time unit shock today, and nothing else happens, how do all variables react today, next period, the period after, and so on?” The **impulse response function (IRF)** is that whole path, period by period, until the effect dies out.

For the VAR(1) without intercept, $z_t = \Theta z_{t-1} + \varepsilon_t$, the slides derive the response after s periods by differentiating:

$$\frac{\partial z_{t+s}}{\partial \varepsilon_t} = \Theta^s, \quad s = 0, 1, 2, \dots \quad (\Theta^0 = I).$$

In words: at $s = 0$ the response matrix is the identity; after one period it is Θ ; after two periods $\Theta \times \Theta$; and so on.

Worked example (the sample-exam type)

Take $\Theta = \begin{pmatrix} -0.1 & 0.2 \\ 0 & 0.2 \end{pmatrix}$, uncorrelated errors. What is the response of y to a unit shock in x ?

$s = 0$: $\Theta^0 = I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$. The response of y to x 's shock is the top-right entry: **0**. Reason: y_t depends only on *last period's* x , and the two errors are uncorrelated, so y cannot react in the same period.

$s = 1$: $\Theta^1 = \Theta$; top-right entry: $\theta_{12} = \mathbf{0.2}$. Next period, y rises by 0.2.

$s = 2$: $(\Theta^2)_{12} = \theta_{11}\theta_{12} + \theta_{12}\theta_{22} = (-0.1)(0.2) + (0.2)(0.2) = \mathbf{0.02}$. The effect fades.

7.7 Granger causality

x **Granger-causes** y if past values of x help predict y , beyond what y 's own past already tells us. In the VAR above: test $H_0 : \theta_{12} = 0$ (with more lags, all of x 's lag coefficients in y 's equation, by an F -test). Careful with the word “causes”: this is about *prediction*, not about one variable truly driving the other. Stock prices often “Granger-cause” GDP simply because markets anticipate the future.

Real-world example: white and black pepper (VAR tutorial)

Monthly price changes of white (r_w) and black (r_b) pepper, 1973–1996, correlation 0.55. Both are the same crop, processed differently.

- **VAR(1) estimates:** last month's black-pepper change is significant in *both* equations

($t = 5.0$ and 4.3); last month's white-pepper change in *neither*. Black leads, white follows.

- **Model choice:** AIC and BIC both prefer VAR(1) over VAR(2); added trends are insignificant (a trend in *price changes* would mean ever-accelerating prices).
- **Granger causality:** black $\rightarrow$ white: $F = 25.3$, $p < 0.0001$ (rejected); white $\rightarrow$ black: $p = 0.21$ (not rejected). One-directional.
- **Impulse response** (Figure 7.1): a black-pepper shock raises white-pepper prices by about 2 percentage points one month later; the effect is gone within about five months.

Plausible story: black pepper is the benchmark, more heavily traded grade, so news about harvests and demand shows up there first. Remember, though, that Granger causality is about prediction — black pepper may simply absorb common news faster. (R's `irf()` additionally “orthogonalises” the shocks and measures them in standard deviations, a refinement beyond the slides; the message is the same.)

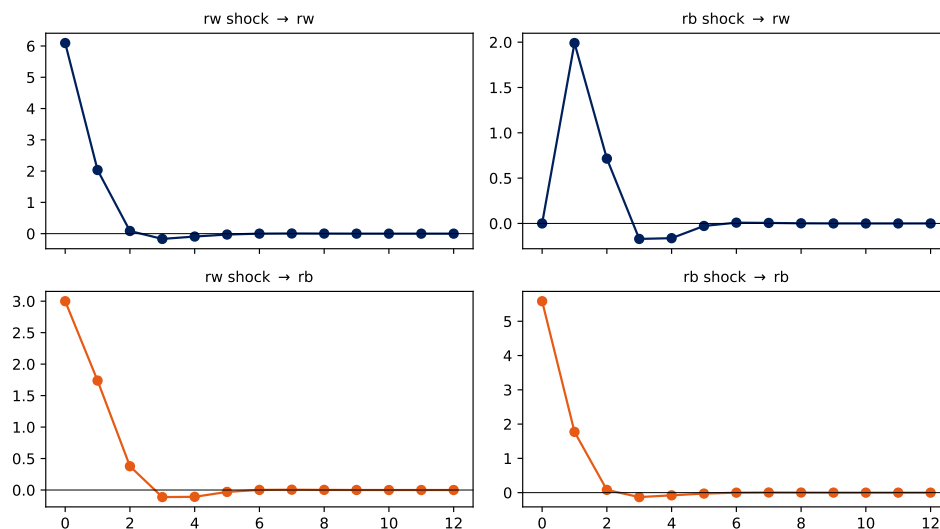


Figure 7.1: Pepper VAR: impulse responses over 12 months.

7.8 The curse of dimensionality

Count the parameters. A VAR with N variables and p lags has, in each of its N equations, one intercept plus $N \times p$ lag coefficients:

$$\text{number of mean parameters} = N(1 + Np) = N + pN^2,$$

plus the variances and covariances of the errors. The number grows with the *square* of the number of variables.

VAR	parameters (mean equations)	with 300 monthly observations...
2 variables, 1 lag	$2 \times 3 = 6$	easy
10 variables, 1 lag	$10 \times 11 = 110$	very noisy
10 variables, 4 lags	$10 \times 41 = 410$	more parameters than observations

The idea in plain words

Every extra variable adds a new coefficient to *every* equation. Parameters grow with N^2 ; data only grow with time. Soon there are too many unknowns for the data to pin down: estimates become imprecise, the model overfits and forecasts badly. That is the **curse of dimensionality**. The remedies are restrictions (structural VARs), penalized regression and factor models — Chapter 9.

7.9 Trends and cointegration in a VAR

If the VAR includes a significant deterministic trend and its coefficients satisfy the stationarity condition, the variables are **trend-stationary**: shocks die out around a trend line. There is no stochastic trend, so a cointegration model (which is designed for $I(1)$ variables sharing a stochastic trend) is *not* appropriate. Keep the trend in the VAR instead.

Chapter 8

Volatility Models: ARCH, GARCH and Friends

This is the largest question in the sample exam (30 points), so we take it slowly.

8.1 Why model volatility at all?

An ARMA model assumes the variance of the shocks is *constant*: every period is equally uncertain. Financial data break this assumption: they have calm periods and turbulent periods (Figure 8.1). And in finance, uncertainty is often *the* quantity of interest: the risk of a portfolio, the price of an option, how much capital a bank must hold.

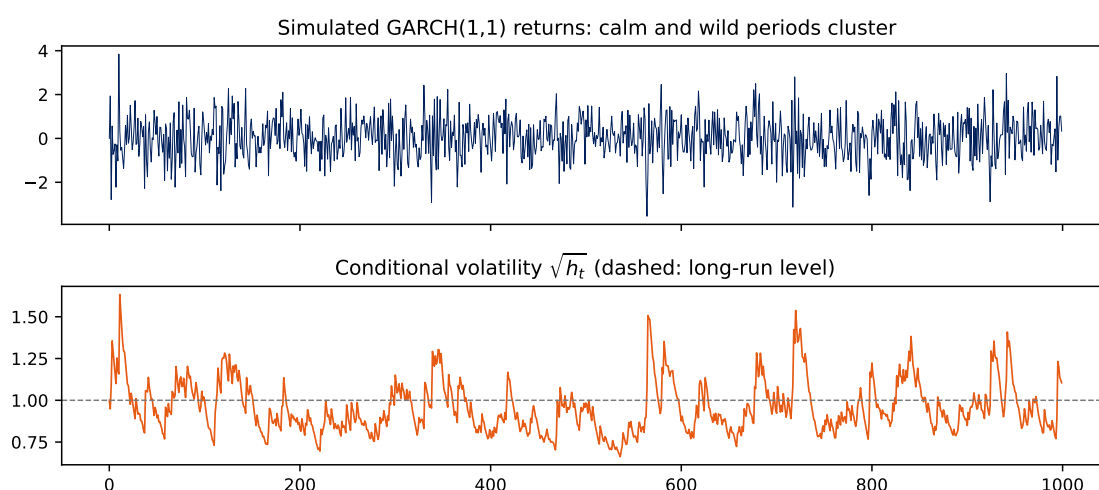


Figure 8.1: Volatility clustering: the model's volatility rises in wild periods and falls in calm ones.

Two risk measures from the slides show why the *tails* and the *variance* matter more than the mean:

- **Value-at-Risk:** $VaR_t(\alpha)$ is the loss threshold such that $\Pr(y_t < VaR_t(\alpha)) = \alpha$. For $\alpha = 1\%$: the return is worse than VaR only one day in a hundred.
- **Expected shortfall:** $ES_t(\alpha) = E(y_t \mid y_t < VaR_t(\alpha))$ — the average loss on those worst days. It says not only *how often*, but *how bad*.

Both carry a subscript t because they change over time — precisely because volatility changes.

Exam corner

“Why use a conditional volatility model instead of ARMA?” Answer: *ARMA models the conditional mean and assumes a constant variance. Many economic and financial series show volatility that changes over time and clusters, and the uncertainty itself is of interest. Example: stock index returns, which are much more volatile in crises (2008) than in calm years; or US GDP growth, much less volatile after 1985 (the Great Moderation).*

8.2 Why not just fix OLS?

We have met two ways in which regression errors misbehave: autocorrelation (cured by ARMA models) and heteroskedasticity (turbulent and calm periods). Under both, OLS remains consistent but its standard errors are wrong. A quick patch is to use **HAC standard errors** — but the patch does not work for every kind of dependence, and it only repairs the inference. If we *model* the changing variance directly, we get better estimates and better forecasts — including forecasts of risk itself.

Real-world example: the mean is hard, the variance is easier

The lecture plots daily returns of the NYSE International 100 index. The returns hover around zero: forecasting the *mean* is hopeless. But the volatility clearly clusters — months of calm, then months of turbulence. Forecasting the *variance* is “relatively more straightforward”. That asymmetry is why finance invests so much in volatility models.

Real-world example: Value-at-Risk and expected shortfall in euros

A trader holds a portfolio worth €1,000,000. The volatility model says tomorrow’s standard deviation is 1.5%, returns are roughly normal with mean zero.

- **1% VaR:** the 1% quantile of a standard normal is -2.33 , so $VaR = -2.33 \times 1.5\% \approx -3.5\%$. One day in a hundred, the loss exceeds about €35,000.
- **1% expected shortfall:** on those worst days, the *average* loss is about -4.0% , roughly €40,000.
- **After a crash** the model may raise volatility to 3%: the VaR doubles to about €70,000. Same portfolio, twice the risk — this is why VaR has a time subscript.

8.3 Conditional versus unconditional variance

The idea in plain words

The **conditional variance** is the variance *given what we know today* — after a crash, tomorrow is expected to be risky. The **unconditional (long-run) variance** is the average level the conditional variance keeps returning to. A series can have a changing conditional variance and still a constant unconditional variance (so it can still be stationary).

Real-world example

Climate versus weather. The average October temperature in Maastricht is about 11°C: that is the *unconditional* expectation. But if today is an unusually warm 20°C, your best forecast for tomorrow is much closer to 20 than to 11: that is the *conditional* expectation, given today’s information. Volatility works the same way: after a crash, tomorrow’s expected volatility is far above the long-run average, and only gradually drifts back.

8.4 The basic measurement problem

With a constant variance, volatility is easy to estimate: $\sigma^2 \approx \frac{1}{T} \sum_{t=1}^T (r_t - E(r_t))^2$, and the larger T , the better. But a *time-varying* variance σ_t^2 has, at each date t , only **one** observation to learn

from. The squared return r_t^2 is an unbiased but very noisy measure (one huge day says little about whether the whole week was risky). The course offers three routes:

1. **measure it** better with intraday data — realized volatility, modelled with HAR;
2. **model it** as a deterministic function of past returns — ARCH and GARCH;
3. treat it as a separate random process — stochastic volatility (presented briefly).

8.5 Measuring volatility: realized volatility and the HAR model

Volatility is not directly observed. Squared returns are a crude, noisy measure. A better one is **realized volatility (RV)**: use many intraday returns (e.g. $N = 79$ five-minute returns per trading day) and compute their spread within each day:

$$RV_t^{(d)} = \sqrt{\frac{1}{N} \sum_{i=1}^N (r_{t,i} - E(r_{t,i}))^2}.$$

Weekly RV is the average of the last 5 daily RVs; monthly RV the average of the last 4 weekly RVs.

The **HAR (heterogeneous autoregressive) model**, or *volatility cascade*, builds on a story about investors. A long-term (monthly) investor cares about monthly risk. A medium-term (weekly) investor cares about weekly and monthly risk. A short-term (daily) investor is exposed to all three. Each forms an expectation from what they see now plus what they expect at the longer horizon:

$$\begin{aligned}\sigma_{t+1m}^{(m)} &= c^{(m)} + \phi^{(m)} RV_t^{(m)} + \varepsilon_{t+1}^{(m)}, \\ \sigma_{t+1w}^{(w)} &= c^{(w)} + \phi^{(w)} RV_t^{(w)} + \gamma^{(w)} E_t[\sigma_{t+1m}^{(m)}] + \varepsilon_{t+1}^{(w)}, \\ \sigma_{t+1d}^{(d)} &= c^{(d)} + \phi^{(d)} RV_t^{(d)} + \gamma^{(d)} E_t[\sigma_{t+1w}^{(w)}] + \varepsilon_{t+1}^{(d)}.\end{aligned}$$

Deriving the cascade by substitution

1. From the monthly equation: $E_t[\sigma_{t+1m}^{(m)}] = c^{(m)} + \phi^{(m)} RV_t^{(m)}$.
2. Put this into the weekly equation: $E_t[\sigma_{t+1w}^{(w)}] = c^{(w)} + \gamma^{(w)} c^{(m)} + \phi^{(w)} RV_t^{(w)} + \gamma^{(w)} \phi^{(m)} RV_t^{(m)}$.
3. Put that into the daily equation and collect terms:

$$\sigma_{t+1d}^{(d)} = c + \beta^{(d)} RV_t^{(d)} + \beta^{(w)} RV_t^{(w)} + \beta^{(m)} RV_t^{(m)} + \varepsilon_{t+1}^{(d)},$$

with $c = c^{(d)} + \gamma^{(d)} c^{(w)} + \gamma^{(d)} \gamma^{(w)} c^{(m)}$, $\beta^{(d)} = \phi^{(d)}$, $\beta^{(w)} = \gamma^{(d)} \phi^{(w)}$, $\beta^{(m)} = \gamma^{(d)} \gamma^{(w)} \phi^{(m)}$.

In words: tomorrow's daily volatility depends on today's daily, weekly and monthly realized volatility — long-horizon risk cascades down to short horizons. The model is a simple regression, easy to estimate if you have intraday data.

8.6 The ARCH(1) model

ARCH (AutoRegressive Conditional Heteroskedasticity) was the first model of changing volatility. Write the return as a shock whose size changes over time:

$$r_t = \varepsilon_t, \quad \varepsilon_t = \sqrt{h_t} v_t, \quad v_t \sim NID(0, 1), \quad h_t = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2.$$

In words: v_t is a standard “unit” shock; $\sqrt{h_t}$ scales it up or down. Today’s variance h_t is large if yesterday’s shock was large.

Why h_t is the conditional variance (slide derivation): given yesterday’s information I_{t-1} , h_t is already known, so $\text{Var}(r_t | I_{t-1}) = h_t \text{Var}(v_t) = h_t \times 1 = h_t$. It changes with t — that is the point.

Real-world example: ARCH(1) after a calm day and after a crash

Daily returns in %, with $h_t = 0.2 + 0.5 \varepsilon_{t-1}^2$.

- After a calm day ($\varepsilon_{t-1} = 0.5$): $h_t = 0.2 + 0.5 \times 0.25 = 0.325$, so tomorrow’s volatility is $\sqrt{0.325} \approx 0.57\%$.
- After a crash day ($\varepsilon_{t-1} = -4$): $h_t = 0.2 + 0.5 \times 16 = 8.2$, so tomorrow’s volatility is $\sqrt{8.2} \approx 2.86\%$ — five times larger.
- Long run: $0.2 / (1 - 0.5) = 0.4$, a long-run volatility of 0.63% .

Note that the crash and a +4% rally would give exactly the same h_t : ARCH ignores the sign.

8.7 The GARCH(1,1) model: the workhorse

GARCH (Generalized ARCH) adds yesterday’s variance to the recipe:

$$h_t = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 h_{t-1}.$$

In words: today’s variance = a base level + a reaction to yesterday’s news (α_1) + a memory of yesterday’s variance (β_1).

Watch out

The slides are not consistent about which Greek letter sits on which term (one slide puts α_1 on h_{t-1} and β_1 on ε_{t-1}^2). Never rely on the letter: **always look at which term the coefficient multiplies**. In this book, α_1 multiplies the squared shock and β_1 the lagged variance.

Because h_{t-1} itself contains ε_{t-2}^2 and h_{t-2} , and so on, GARCH(1,1) secretly lets *all* past squared shocks matter, with weights that fade geometrically. That is why one GARCH(1,1) usually captures the volatility dynamics as well as a long ARCH model with many lags.

Long-run (unconditional) variance of GARCH(1,1)

1. Take expectations: $E(h_t) = \alpha_0 + \alpha_1 E(\varepsilon_{t-1}^2) + \beta_1 E(h_{t-1})$.

2. Replace $E(\varepsilon_{t-1}^2)$: since $\varepsilon_{t-1}^2 = h_{t-1}v_{t-1}^2$, and v_{t-1} is independent of h_{t-1} (which depends only on earlier shocks), $E(\varepsilon_{t-1}^2) = E(h_{t-1})E(v_{t-1}^2) = E(h_{t-1}) \times 1$.

3. Use stationarity: $E(h_t) = E(h_{t-1})$. So $E(h) = \alpha_0 + \alpha_1 E(h) + \beta_1 E(h)$.

4. Solve:
$$E(h_t) = \frac{\alpha_0}{1 - \alpha_1 - \beta_1}$$

Conditions for sensible variances (slide):

- *Short-run (conditional) variance* must be positive every period: sufficient conditions $\alpha_0 > 0$, $\alpha_1 \geq 0$, $\beta_1 \geq 0$.
- *Long-run (unconditional) variance* must be finite and positive: $\alpha_0 > 0$ and $\alpha_1 + \beta_1 < 1$.

Persistence. The sum $\alpha_1 + \beta_1$ tells how long a volatility shock lasts; its *half-life* is $\ln 0.5 / \ln(\alpha_1 + \beta_1)$ periods. If $\alpha_1 + \beta_1 = 1$, shocks to volatility never die out (“IGARCH”) and the long-run variance does not exist.

Worked example: MSCI Switzerland (GARCH tutorial)

Estimates: $\alpha_0 = 0.023$, $\alpha_1 = 0.105$, $\beta_1 = 0.877$ (daily % returns).

Positivity: all three positive ✓. **Long run:** $\alpha_1 + \beta_1 = 0.982 < 1$ ✓.

Long-run variance: $0.023 / (1 - 0.982) \approx 1.32$, so long-run volatility $\sqrt{1.32} \approx 1.15\%$ per day.

Half-life: $\ln 0.5 / \ln 0.982 \approx 39$ days — after a shock, risk stays high for about two months.

Reading: about 10% of yesterday’s squared surprise feeds into today’s variance (news); about 88% of yesterday’s variance carries over (memory).

What standard GARCH can and cannot do. It captures *volatility clustering* and produces *fat tails* (mixing calm and wild days creates extra extreme values). It cannot capture the *leverage effect*, because it uses the *squared* shock — a -2% day and a $+2\%$ day raise tomorrow’s variance by the same amount. And volatility does not affect the mean return unless you add it (ARCH-in-mean, below).

More lags: GARCH(p, q). The general version allows q lagged squared shocks and p lagged variances: $h_t = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \cdots + \alpha_q \varepsilon_{t-q}^2 + \beta_1 h_{t-1} + \cdots + \beta_p h_{t-p}$. In practice GARCH(1,1) is usually enough.

Real-world example: reading history from a GARCH model

Figure 8.2 shows the GARCH tutorial’s estimated volatility for the Swiss stock market. Every peak is a named crisis: 1998 (Asian and Russian crises, the LTCM hedge-fund collapse), 2001–2003 (dot-com bust, 9/11), 2008 (the maximum, 5.07% per day on **14 October 2008**, a month after Lehman Brothers failed) and 2011 (euro debt crisis). In calm 2005, volatility averaged only 0.68%. And the peaks are *plateaus*, not spikes: with persistence 0.982, high risk lingers for weeks — the 39-day half-life you can see with your own eyes.

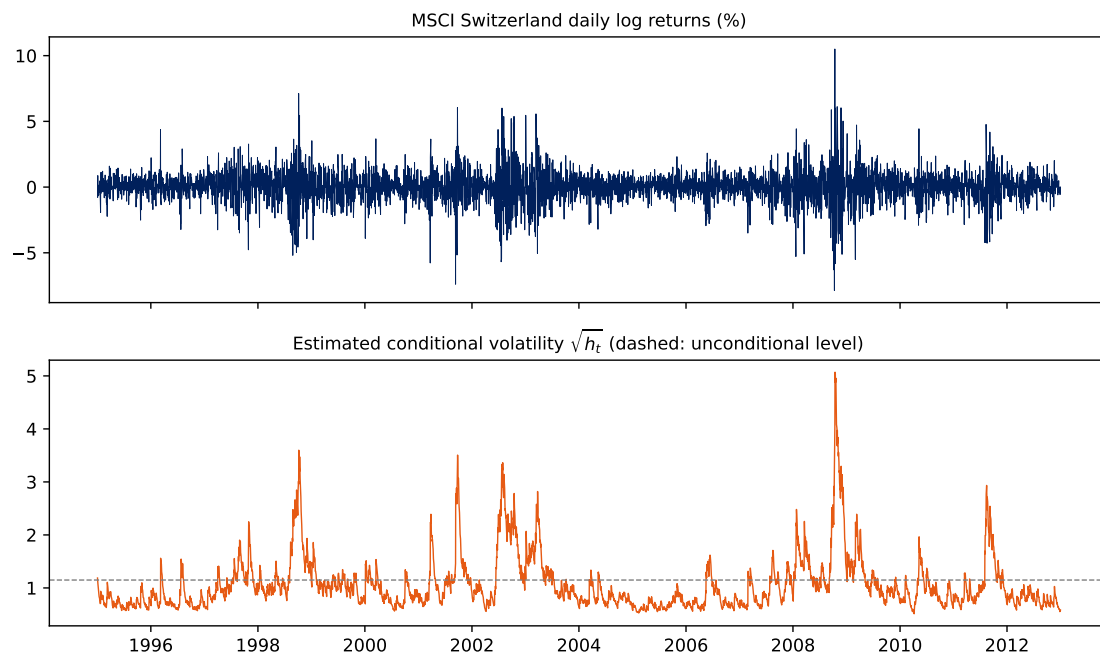


Figure 8.2: MSCI Switzerland, 1995–2012: daily returns and GARCH(1,1) conditional volatility.

Real-world example: the leverage effect

Why would bad news raise volatility more than good news? Take a firm financed with €50 of equity and €50 of debt. If its share value falls to €25, debt is now two-thirds of the firm instead of half: the firm is more *leveraged*, so its shares become riskier. Add panic selling and margin calls in falling markets, and you get the typical pattern: volatility jumps when prices fall (2008, March 2020), much less when they rise.

8.8 When variables enter the variance: the positivity problem

The sample exam adds an explanatory variable to an ARCH(2) model (using η_t for the shock and σ_t^2 for the variance — same ideas, different letters):

$$y_t = \mu + \eta_t, \quad \eta_t = \sigma_t \epsilon_t, \quad \epsilon_t \sim NID(0, 1), \quad \sigma_t^2 = \alpha_0 + \alpha_1 \eta_{t-1}^2 + \alpha_2 \eta_{t-2}^2 + \alpha_3 x_{t-1}.$$

The problem with a negative coefficient. The squared terms can never be negative, but x_{t-1} can be any size. If $\alpha_3 = -5$ and $\alpha_0 = 0.1$, then any x_{t-1} above 0.02 (with small recent shocks) gives $\sigma_t^2 < 0$ — a negative variance, which is impossible. The model breaks down, and the likelihood cannot even be evaluated.

Unconditional mean. $E(y_t) = \mu + E(\eta_t) = \mu + E(\sigma_t)E(\epsilon_t) = \mu$, because ϵ_t has mean zero and is independent of σ_t . So $E(y_t) = \mu$; the variance parameters do not matter for the mean.

Unconditional variance. Take expectations of the variance equation and use $E(\eta_{t-k}^2) =$

$E(\sigma_{t-k}^2)E(\epsilon_{t-k}^2) = E(\sigma^2)$ (stationarity):

$$E(\sigma^2) = \alpha_0 + \alpha_1 E(\sigma^2) + \alpha_2 E(\sigma^2) + \alpha_3 \mu_x \implies \boxed{E(\sigma_t^2) = \frac{\alpha_0 + \alpha_3 \mu_x}{1 - \alpha_1 - \alpha_2}}.$$

This is also the unconditional variance of returns, since $\text{Var}(y_t) = E(\eta_t^2) = E(\sigma_t^2)$. It requires $\alpha_1 + \alpha_2 < 1$ and a positive numerator.

8.9 Allowing for the leverage effect: TARCH and EGARCH

TARCH (threshold GARCH) adds an extra reaction to bad news only:

$$h_t = \alpha_0 + \alpha_1 \epsilon_{t-1}^2 + \lambda_1 I\{\epsilon_{t-1} < 0\} \epsilon_{t-1}^2 + \beta_1 h_{t-1}.$$

$I\{\cdot\}$ equals 1 after a negative shock and 0 otherwise. A negative shock adds $(\alpha_1 + \lambda_1)\epsilon^2$, a positive one only $\alpha_1\epsilon^2$. So $\lambda_1 > 0$ is the leverage effect. Its long-run variance is $\alpha_0/(1 - \alpha_1 - \beta_1 - \frac{1}{2}\lambda_1)$ (half of all shocks are negative, for a symmetric distribution).

EGARCH (exponential GARCH) models the *logarithm* of the variance:

$$\log h_t = \alpha_0 + \alpha_1 \left| \frac{\epsilon_{t-1}}{\sqrt{h_{t-1}}} \right| + \lambda_1 \frac{\epsilon_{t-1}}{\sqrt{h_{t-1}}} + \beta_1 \log h_{t-1}.$$

The idea in plain words

The log trick: $\log h_t$ may be any number, positive or negative — and then $h_t = \exp(\log h_t)$ is **always positive**, whatever the parameter values. So EGARCH needs **no positivity restrictions at all**. That is exactly how it rescues the sample-exam model with $\alpha_3 = -5$: put x_{t-1} into the log-variance equation, and a negative coefficient is no longer a problem.

Leverage sits in λ_1 , the coefficient on the *signed* shock (the absolute-value term carries the size effect). $\lambda_1 < 0$ means bad news raises volatility more than good news.

Real-world example: TARCH and EGARCH in numbers

TARCH with $\alpha_0 = 0.05$, $\alpha_1 = 0.05$, $\lambda_1 = 0.10$, $\beta_1 = 0.85$: a -2% day adds $(0.05 + 0.10) \times 4 = 0.60$ to tomorrow's variance; a $+2\%$ day adds only $0.05 \times 4 = 0.20$ — bad news has three times the impact. Long-run variance: $0.05/(1 - 0.05 - 0.85 - 0.05) = 1$.

EGARCH keeps everything positive: if the right-hand side of the log-variance equation equals -1.5 , then $h_t = e^{-1.5} \approx 0.22$; even at -3 , $h_t = e^{-3} \approx 0.05$. A negative number on the log scale is simply a small, still positive, variance.

8.10 Volatility in the mean: ARCH-M

To let risk move expected returns, put the variance into the mean equation:

$$y_t = \mu + \delta \sigma_{t-1}^2 + \eta_t.$$

δ measures the effect of last period's volatility on today's mean return. A positive δ is a risk premium: investors want higher returns after riskier periods.

8.11 Stochastic volatility

In GARCH, volatility is a fixed function of past returns. In the **stochastic volatility (SV)** model, volatility has *its own* random shock:

$$r_t = \epsilon_t \exp(h_t/2), \quad h_{t+1} = \mu + \phi(h_t - \mu) + \eta_t,$$

with $\epsilon_t \sim NID(0,1)$ and $\eta_t \sim NID(0, \sigma_\eta^2)$. Here h_t is the *log* of the variance, so (log trick again) the variance $\exp(h_t)$ is always positive, and the volatility equation is a simple linear AR(1). With $|\phi| < 1$ volatility is stationary; with $\phi = 1$ it behaves like a random walk. Because volatility is not observed, we cannot run a unit-root test on it. SV models also produce fat tails, and multivariate versions exist.

8.12 Evaluating volatility forecasts

For the *mean*, compare forecasts $\hat{r}_f$ with actual returns using MSFE and MAFE, as in Chapter 4. For the *volatility*, there is a snag: true volatility is never observed. The simplest fix compares the variance forecasts $\hat{h}_f$ with squared returns r_f^2 ; a better fix uses realized volatility RV_f from intraday data:

$$MSFE = \frac{1}{F} \sum_{f=1}^F (r_f^2 - \hat{h}_f)^2 \quad \text{or} \quad \frac{1}{F} \sum_{f=1}^F (RV_f^2 - \hat{h}_f)^2.$$

Big Data: Factor Models, PCA, Ridge and Lasso

9.1 The problem: too many variables

Suppose you observe N series over T periods (a $T \times N$ data matrix). When N is close to T — or even larger — the data are **high-dimensional**. Examples: 100 stocks over one year ($N = 100$, $T = 252$); or the sample exam's 200 explanatory variables for only 56 months of sales.

OLS breaks down in this setting. With more variables than observations ($N > T$) it has no unique solution at all; with N close to T it fits the noise of the sample, giving huge, unstable coefficients and terrible forecasts.

The idea in plain words

Including every potentially relevant variable is a bad idea. We must **reduce the dimension**. There are two strategies:

Aggregation — summarise many variables in a few combinations (factor models, PCA). Useful when *all* variables carry some information.

Subset selection — keep only the strong variables (Lasso). Useful when only a *few* variables matter.

Watch out

All methods in this chapter assume **stationary** data (the slides say so explicitly). Applying PCA or penalized regression to trending, wandering series would mostly pick up the common trends — spurious structure. Test for unit roots and difference first.

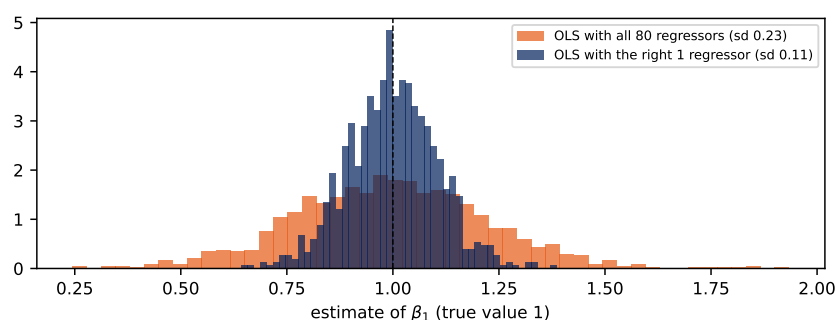


Figure 9.1: The slides' illustration, simulated: $T = 100$, $N = 80$ candidate regressors, but only x_1 truly matters.

Real-world example: what too many regressors do

The slides simulate $T = 100$ observations of $N = 80$ variables, where in truth only the first matters: $y_t = \alpha + 1 \cdot x_{1,t} + \epsilon_t$. Figure 9.1 repeats the experiment 1,000 times. OLS with all 80 regressors still centres on the true value 1, but its estimates are **more than twice as dispersed** (standard deviation 0.23 against 0.10 for the correct small model). Every useless

variable costs precision. In reality we do not know which variables are the useless ones — that is the problem this chapter solves.

Sparsity. Penalized regression often assumes the true β is *sparse*: of the N coefficients, only s are non-zero. Alternatively, all may be non-zero, but only a few are needed for good predictions.

9.2 Factor models

A factor model says that many observed series are driven by a few hidden common factors:

$$\underbrace{x_t}_{N \times 1} = \underbrace{\Lambda}_{N \times r} \underbrace{f_t}_{r \times 1} + \underbrace{\epsilon_t}_{N \times 1}, \quad r \ll N.$$

In words: the N series are mostly explained by r common factors f_t ; the *loadings* in Λ say how strongly each series responds to each factor; ϵ_t is what is specific to each series.

Example: 100 stock returns are largely driven by one “market” factor plus a few sector factors. Sometimes factors are observed (e.g. the market return); usually they are not, and we estimate them by principal components.

Why factors help a regression. Suppose we want $y_t = \beta' x_t + \epsilon_t$ with a huge vector x_t generated by a factor model $x_t = \Lambda f_t + u_t$. Substitute:

$$y_t = \beta' (\Lambda f_t + u_t) + \epsilon_t = \tilde{\beta}' f_t + \tilde{\epsilon}_t, \quad \tilde{\beta}' = \beta' \Lambda.$$

In words: instead of estimating N coefficients, we estimate only r — one per factor.

Real-world example: observed factors — Fama and French

Sometimes the factors are directly observed. The famous Fama–French three-factor model explains the excess return of any stock i by three observed series:

$$r_{i,t} = \beta_{i,1} MKT_t + \beta_{i,2} SMB_t + \beta_{i,3} HML_t + \epsilon_{i,t},$$

where MKT is the market’s excess return, SMB (“small minus big”) the return of small firms over large ones, and HML (“high minus low”) the return of value stocks over growth stocks. Thousands of stocks, but each needs only three loadings.

9.3 Principal components analysis (PCA)

The idea in plain words

PCA looks for the single combination of all variables that captures as much of their joint variation as possible: that is the **first principal component**. Then it finds the next combination that captures as much of the *remaining* variation as possible, while being uncorrelated with the first — and so on.

The technical engine is the *eigenvalues and eigenvectors* of the covariance (or correlation) matrix of the data. You only need to know what they mean:

- each **eigenvalue** is the amount of variation explained by one component; divided by their total, it gives the *proportion of variance explained*;
- each **eigenvector** holds the weights (loadings) that build that component from the original variables;
- components are **uncorrelated**, so each explains a unique part of the data — a few components can explain a lot.

A tiny example. Two standardized series with correlation $\rho = 0.8$. The first component is (proportional to) their *average*, and it explains $(1 + \rho)/2 = 90\%$ of the total variation. The second is their *difference* and explains the remaining 10%. One component nearly summarises both series.

Interpretation is tricky, for two reasons (slides): each component mixes *all* variables, and the factors are not uniquely defined — $\lambda_i f_t = (\lambda_i/2)(2f_t)$, so you can rescale a factor and its loading without changing anything. Standardize variables measured in different units before PCA, otherwise the variable with the largest variance dominates purely because of its units.

A short detour: eigenvalues and eigenvectors. For a square matrix A , if $Av_k = \lambda_k v_k$, then v_k is an *eigenvector* and λ_k its *eigenvalue*: multiplying v_k by A only stretches it, without changing its direction. For a symmetric matrix (like a covariance matrix) the eigenvalues are real numbers and the eigenvectors are perpendicular to each other ($v_j'v_k = 0$), and they can be scaled to length one.

How PCA estimates the factors. We choose Λ and the factors to make the common part Λf_t as close as possible to the data, minimizing $\sum_t (x_t - \Lambda f_t)'(x_t - \Lambda f_t)$. Because a factor and its loading can be rescaled freely (the identification problem above), we add the constraint $\Lambda'\Lambda = I_r$. The solution: $\hat{\Lambda}$ consists of the first r eigenvectors of the sample covariance matrix $\frac{1}{T}X'X$, and the factors are $\hat{f}_t = \hat{\Lambda}'x_t$.

Real-world examples: what principal components find

Gold and silver (lecture, correlation 0.85): the first component is essentially their average, a “precious-metals factor”, explaining $(1 + 0.85)/2 \approx 92\%$ of their joint variation.

The yield curve. Apply PCA to government bond yields of, say, ten maturities. Almost always, PC1 is a *level* factor (all yields move up or down together), PC2 a *slope* factor (short and long yields move in opposite directions), and PC3 a *curvature* factor. Together these three explain about 99% of daily yield-curve movements, so a bond trader can manage the risk of ten maturities by watching three numbers.

9.4 How many components?

1. **Look at the eigenvalues (scree plot).** Find the *elbow* — the point after which extra components add little. Keep the components up to the elbow (Figure 9.2: two).

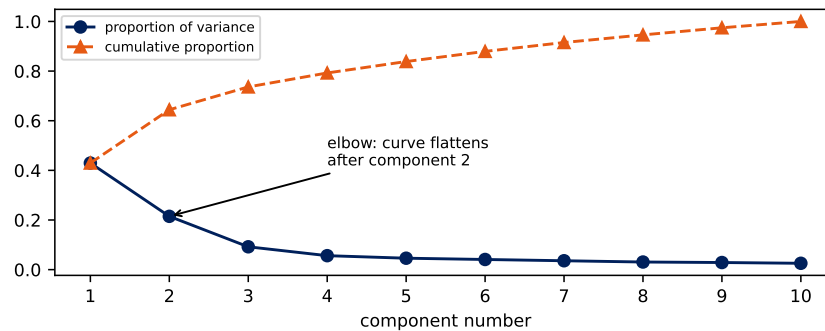


Figure 9.2: A scree plot: keep the components before the curve flattens.

2. **Cumulative proportion:** keep enough components to explain a chosen share of the variation (e.g. 80%).
3. **Information criteria.**
4. **Forecast performance**, if the goal is forecasting.

Visual inspection is somewhat subjective, so it is good practice to mention more than one criterion.

Forecasting with factors: estimate the factors $\hat{f}_t$ by PCA, then regress the target on them (and possibly its own lags), $y_{t+h} = \alpha + \beta' \hat{f}_t + \phi y_t + \epsilon_t$, and forecast with the estimated coefficients.

9.5 Subset selection and penalized regression

Ideally we would try every possible subset of variables and pick the best. But with N variables there are 2^N subsets: $N = 10$ already gives 1,024 models, $N = 200$ an astronomical number. Penalized regression does estimation and selection *at the same time*.

The idea: estimate the full model, but restrict the total size of the coefficients. Equivalently, minimize the usual sum of squared errors *plus a penalty*:

$$\text{Ridge: } \sum_t (y_t - \beta' x_t)^2 + \lambda \sum_j \beta_j^2 \qquad \text{Lasso: } \sum_t (y_t - \beta' x_t)^2 + \lambda \sum_j |\beta_j|.$$

In words: fit the data well, but pay a price λ for large coefficients. $\lambda = 0$ gives OLS; a huge λ pushes all coefficients towards zero.

Ridge with one variable (slide derivation)

Minimize $\sum_t (y_t - \beta x_t)^2 + \lambda \beta^2$. Set the derivative to zero: $-2 \sum_t x_t (y_t - \hat{\beta} x_t) + 2\lambda \hat{\beta} = 0$, so

$$\hat{\beta}_{\text{ridge}} = \frac{\sum_t x_t y_t}{\sum_t x_t^2 + \lambda}.$$

Compare with OLS, $\sum x_t y_t / \sum x_t^2$: the λ in the denominator **shrinks** the estimate towards zero — but never exactly to zero.

Ridge versus Lasso. Both allow you to estimate models with many variables. Ridge shrinks all coefficients but keeps every variable. Lasso can set coefficients **exactly to zero**, so it also *selects* variables. (Lasso's absolute value has no derivative at zero, so it is computed with numerical methods.) Figure 9.3 shows why: the error contours touch the Lasso's diamond-shaped budget at a corner, where one coefficient is zero; the Ridge's round budget has no corners.

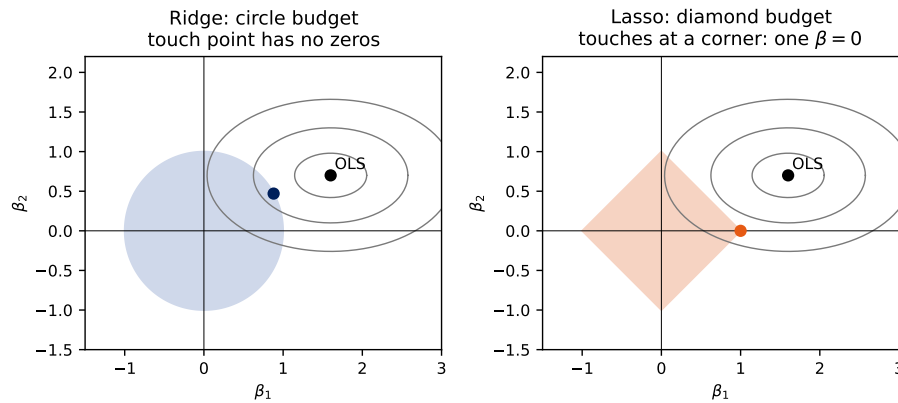


Figure 9.3: Why Lasso produces exact zeros and Ridge does not.

Elastic net combines both penalties ($\lambda_1 \sum |\beta_j| + \lambda_2 \sum \beta_j^2$): it selects like Lasso while handling strongly correlated variables more gracefully, like Ridge.

Two equivalent formulations. The slides first write penalized regression as a *budget*: minimize the squared errors subject to $\|\beta\|_q \leq K$, where $\|\beta\|_q = (\sum_j |\beta_j|^q)^{1/q}$ (the L_q -norm; $q = 1$ gives Lasso, $q = 2$ Ridge). For estimation it is easier to write it as a *penalty*: minimize squared errors $+ \lambda \|\beta\|$. A result called *Lagrangian duality* guarantees the two are the same problem: each penalty λ corresponds to some budget K . A tight budget is a large λ .

Real-world example: watching Ridge shrink

Suppose $\sum_t x_t y_t = 50$ and $\sum_t x_t^2 = 100$. OLS gives $50/100 = 0.5$. Ridge gives $50/(100 + \lambda)$: with $\lambda = 25$ the estimate is 0.4; with $\lambda = 100$ it is 0.25; as λ grows it approaches zero but never reaches it. Lasso, by contrast, would set the coefficient to exactly zero once λ is large enough.

In R: the `glmnet` package minimizes $\frac{1}{2T} \sum_t (y_t - \beta' x_t)^2 + \lambda \left[\frac{1-\alpha}{2} \sum_j \beta_j^2 + \alpha \sum_j |\beta_j| \right]$. Its α (a different α from the rest of the book!) chooses the method: $\alpha = 1$ is Lasso, $\alpha = 0$ is Ridge, and anything in between is the elastic net.

9.6 Choosing the penalty λ

Higher λ means more shrinkage and sparser models. The slides give two approaches. Both start from a grid of λ values between 0 and λ_{max} (the smallest value that sets all coefficients to zero):

- **BIC:** estimate the model for each λ on the grid and pick the one with the lowest BIC.
- **Cross-validation:** split the data into K parts; estimate on $K - 1$ parts, predict the left-out

part, repeat for every part, and pick the λ with the smallest prediction error. (With time series, a split that respects time order — always estimating on the past and predicting the future — is the safer choice.)

The details of both routes.

- **BIC:** for each λ_i on the grid compute $BIC(\lambda_i) = \ln \left(\frac{1}{T} \sum_t (y_t - \hat{\beta}(\lambda_i)'x_t)^2 \right) + s(\lambda_i) \frac{\ln T}{T}$, where $s(\lambda_i)$ is the number of non-zero coefficients. Choose the λ with the lowest BIC; no re-estimation is needed.
- **Cross-validation:** (1) split the data into K parts; (2) for each λ_i , estimate on $K - 1$ parts and predict the left-out part, rotating through all K parts; (3) average the prediction errors (e.g. MSFE); (4) choose the λ with the lowest average error; (5) re-estimate the model on the *full* sample with that λ .

Forecasting is then simple: estimate $y_{t+h} = \beta'x_t + \epsilon_t$ by penalized regression and forecast $\hat{y}_{T+h} = \hat{\beta}'x_T$. You can also estimate a whole VAR equation by equation with a penalty — a direct answer to the curse of dimensionality of Chapter 7.

Real-world example: forecasting sales with 200 indicators

In the sample exam's setting (56 months, 200 candidate indicators), Lasso might keep only a handful — perhaps the product's own price, a consumer-confidence index and one exchange rate — and set the other coefficients to zero. Ridge would keep all 200 but shrink them, which tends to work better if many indicators each carry a little information. A factor model would summarise the 200 indicators into a few factors ("the business cycle", "the currency") and regress sales on those.

9.7 Which method when?

Ridge	many variables, each carrying a little information, often highly correlated; you want stable predictions, not a short list
Lasso	you believe only a few variables matter and want to know which ones
Elastic net	both: groups of correlated variables, of which only some matter
PCA / factor model	many series move together because of a few common drivers; you want to summarise them

Exam corner

"Why Ridge instead of OLS with 200 variables and 56 observations?" *With more variables than observations, OLS has no unique solution and would fit the sample perfectly but forecast terribly. Ridge adds a penalty on the size of the coefficients, which makes the estimation possible and reduces the variance of the estimates by shrinking them towards zero. It is especially suitable when many explanatory variables are correlated, as macroeconomic indicators typically are.*

Chapter 10

The Sample Exam, Fully Solved

Each answer below is written the way the exam asks: in full sentences, with the reasoning shown and assumptions stated. Read the question, try it yourself, then compare.

Question 1 (30 points): ARCH(2) with an explanatory variable

Model: $y_t = \mu + \eta_t$, $\eta_t = \sigma_t \epsilon_t$, $\epsilon_t \sim NID(0, 1)$, $\sigma_t^2 = \alpha_0 + \alpha_1 \eta_{t-1}^2 + \alpha_2 \eta_{t-2}^2 + \alpha_3 x_{t-1}$, with x_t stationary.

(a) Why a conditional volatility model instead of ARMA?

An ARMA model describes the conditional *mean* and assumes the variance of the errors is constant over time. Many economic and financial series, however, show variance that changes over time and clusters: turbulent periods follow turbulent periods. A conditional volatility model lets the variance depend on past information, which matters because uncertainty itself is often the object of interest (risk management, Value-at-Risk, option pricing). Example: stock index returns are far more volatile during crises such as 2008 than in calm years; similarly, US GDP growth was much less volatile after 1985 (the Great Moderation).

(b) Why are $\alpha_0 = 0.1$, $\alpha_1 = 0.01$, $\alpha_2 = 0.02$, $\alpha_3 = -5$ a problem?

A variance must be positive. The squared terms are never negative, but $\alpha_3 x_{t-1}$ with $\alpha_3 = -5$ is negative whenever $x_{t-1} > 0$, and x_{t-1} is not bounded. For instance, with $x_{t-1} = 0.1$ and small recent shocks, $\sigma_t^2 \approx 0.1 - 0.5 = -0.4 < 0$: a negative variance, which is impossible, so the model (and its likelihood) breaks down. The sufficient positivity conditions $\alpha_i \geq 0$ (with $\alpha_0 > 0$) are violated by α_3 . (Note that $\alpha_1 + \alpha_2 = 0.03 < 1$ is fine; the problem is only positivity.)

(c) Can EGARCH solve this? Write it down.

Yes. EGARCH models the *logarithm* of the variance, so the variance $\sigma_t^2 = \exp(\cdot)$ is positive for any parameter values, including $\alpha_3 = -5$. A suitable model is

$$\log \sigma_t^2 = \alpha_0 + \alpha_1 |\epsilon_{t-1}| + \alpha_2 |\epsilon_{t-2}| + \alpha_3 x_{t-1}, \quad \epsilon_{t-k} = \eta_{t-k} / \sigma_{t-k},$$

optionally adding $\lambda_1 \epsilon_{t-1}$ (leverage) and $\beta_1 \log \sigma_{t-1}^2$ (persistence). No parameter restrictions are needed for positivity.

(d) Unconditional expectation of y_t .

$E(y_t) = \mu + E(\eta_t) = \mu + E(\sigma_t \epsilon_t) = \mu + E(\sigma_t)E(\epsilon_t) = \mu$, because ϵ_t has mean zero and is independent of σ_t (which depends only on past information). So $E(y_t) = \mu$; it does not depend on $\alpha_0, \dots, \alpha_3$ or μ_x .

(e) Unconditional variance $E(\sigma_t^2)$.

Take expectations of the variance equation: $E(\sigma_t^2) = \alpha_0 + \alpha_1 E(\eta_{t-1}^2) + \alpha_2 E(\eta_{t-2}^2) + \alpha_3 E(x_{t-1})$. Since $\eta_{t-k}^2 = \sigma_{t-k}^2 \epsilon_{t-k}^2$ with ϵ independent of σ and $E(\epsilon^2) = 1$, we get $E(\eta_{t-k}^2) = E(\sigma_{t-k}^2)$, which

under stationarity equals $E(\sigma_t^2)$. Also $E(x_{t-1}) = \mu_x$. Hence

$$E(\sigma_t^2) = \frac{\alpha_0 + \alpha_3 \mu_x}{1 - \alpha_1 - \alpha_2},$$

which requires $\alpha_1 + \alpha_2 < 1$ and $\alpha_0 + \alpha_3 \mu_x > 0$. This is also the unconditional variance of returns, since $\text{Var}(y_t) = E(\eta_t^2) = E(\sigma_t^2)$. (With the numbers of (b), it is positive only if $\mu_x < 0.02$ — the same problem again.)

(f) ARCH effects in the mean.

Add last period's volatility to the mean equation: $y_t = \mu + \delta \sigma_{t-1}^2 + \eta_t$, with the variance equation unchanged. The parameter δ measures the effect of last period's volatility on the mean return; $\delta > 0$ would indicate a risk premium.

Question 2 (40 points): the baseline regression

Model: $y_t = \alpha_0 + \alpha_1 y_{t-1} + \beta_1 x_t + \beta_2 \text{trend}_t + \epsilon_t$, $\epsilon_t \sim NID(0, \sigma^2)$, x_t stationary, $\text{cov}(y_{t-1}, \epsilon_t) = \text{cov}(x_t, \epsilon_t) = 0$.

(a) If $|\alpha_1| < 1$, is y_t stationary?

Not necessarily. $|\alpha_1| < 1$ guarantees that shocks die out (no unit root), but the term $\beta_2 \text{trend}_t$ makes the mean of y_t change over time whenever $\beta_2 \neq 0$. Then y_t is *trend-stationary*, not stationary. y_t is stationary only if, in addition, $\beta_2 = 0$ (and x_t is stationary, as assumed).

(b) Dickey–Fuller tests say x_t is not stationary. Implication for y_t ?

If x_t has a unit root and $\beta_1 \neq 0$, then y_t inherits the stochastic trend: adding stationary terms to a non-stationary component cannot remove it, so y_t is non-stationary as well, even though $|\alpha_1| < 1$. The usual OLS inference on the model is then unreliable (risk of spurious regression), unless y_t and x_t are cointegrated; otherwise we should difference the data.

(c) Long-run mean with $\beta_2 = 0$.

Take expectations and use stationarity, $E(y_t) = E(y_{t-1}) = \mu_y$, with $E(x_t) = \mu_x$ and $E(\epsilon_t) = 0$: $\mu_y = \alpha_0 + \alpha_1 \mu_y + \beta_1 \mu_x$, so

$$\mu_y = \frac{\alpha_0 + \beta_1 \mu_x}{1 - \alpha_1}.$$

(d) Long-run variance with $\beta_2 = 0$.

Take variances; the exogeneity conditions remove all covariances with ϵ_t :

$$\text{Var}(y_t) = \alpha_1^2 \text{Var}(y_{t-1}) + \beta_1^2 \text{Var}(x_t) + \sigma^2 + 2\alpha_1 \beta_1 \text{cov}(y_{t-1}, x_t).$$

Assuming $\text{cov}(y_{t-1}, x_t) = 0$ (e.g. x_t is not related to past y) and using stationarity, $\text{Var}(y_t) = \text{Var}(y_{t-1})$:

$$\text{Var}(y_t) = \frac{\beta_1^2 \sigma_x^2 + \sigma^2}{1 - \alpha_1^2}, \quad \sigma_x^2 = \text{Var}(x_t).$$

(e) Lunch effect on returns.

Define $D_t = 1$ if observation t lies between 12:00 and 13:00, and $D_t = 0$ otherwise. Extended model: $y_t = \alpha_0 + \alpha_1 y_{t-1} + \beta_1 x_t + \beta_2 \text{trend}_t + \gamma D_t + \epsilon_t$. The parameter γ reflects the lunch

effect; smaller lunch-time returns imply $\gamma < 0$, testable with a t -test of $H_0 : \gamma = 0$.

(f) Lunch effect on volatility.

Let the error variance depend on the lunch dummy: $\epsilon_t = \sigma_t v_t$, $v_t \sim NID(0, 1)$, with $\sigma_t^2 = \exp(\omega + \lambda D_t)$, keeping the mean equation as in the baseline model. The parameter λ reflects the lunch effect on volatility; less volatile lunch times imply $\lambda < 0$. (The exponential keeps the variance positive; alternatively $\sigma_t^2 = \omega + \lambda D_t$ with $\omega > 0$ and $\omega + \lambda > 0$.)

(g) Expected α_1 and β_2 under the efficient market hypothesis.

Under the efficient market hypothesis, past returns contain no information about future returns, so $\alpha_1 = 0$. Returns should also not drift systematically over time — a deterministic trend would be a predictable source of profit — so $\beta_2 = 0$.

(h) Four methods to assess whether the trend is necessary.

1. *t*-test of $H_0 : \beta_2 = 0$: rejection (e.g. $p < 0.05$) indicates the trend is needed.
2. *Information criteria*: estimate the model with and without the trend; a lower AIC/BIC with the trend indicates it is needed.
3. *Out-of-sample forecast comparison*: a lower MSFE/MAFE for the model with the trend indicates it is needed.
4. *Dickey–Fuller regression with trend* (or a plot of the data): a significant trend coefficient β in $\Delta y_t = \alpha + \beta t + \delta y_{t-1} + \epsilon_t$, or a clear upward/downward drift in the plot, indicates a deterministic trend.

(An F -, Wald or likelihood-ratio test comparing both models is also acceptable.)

Question 3 (15 points): 200 variables, 56 months

Model: $\text{Sales}_t = \beta_0 + \beta' x_t + \epsilon_t$, $t = 1, \dots, 56$, with x_t a 200×1 vector.

(a) Why Ridge instead of OLS?

There are 200 explanatory variables but only 56 observations. With more parameters than observations, OLS has no unique solution (the matrix $X'X$ cannot be inverted) and would fit the sample perfectly while forecasting very badly. Ridge regression adds a penalty $\lambda \sum_j \beta_j^2$ on the size of the coefficients. This makes estimation possible, shrinks the coefficients towards zero and reduces their variance, giving more stable estimates and forecasts — especially useful because macroeconomic and exchange-rate indicators are typically highly correlated. (Ridge keeps all variables; it does not select a subset.)

(b) How many principal components (Figure 1)?

The proportion of variance explained drops sharply from the first component (about 0.2) to the second (well below 0.1) and is nearly flat afterwards; the scree plot has its *elbow* at two components. By the elbow rule, **two principal components** should be used. I note that two components explain only about 30% of the variation in the cumulative curve; if the goal is forecasting, the choice can be verified with information criteria or out-of-sample forecast performance.

(c) Is PCA a good method for these data (Figure 3)?

The sales and price series in Figure 3 wander with long, persistent swings and do not return to a fixed level; they look non-stationary (stochastic trends). PCA, like the other methods in this course, assumes stationary data: with trending series the correlations are dominated by common trends, so the components would mostly capture spurious co-movement rather than meaningful structure. So PCA should not be applied to the levels directly. I would first test for unit roots (ADF) and difference or otherwise transform the series to stationarity, and then apply PCA. (Figure 2 also shows blocks of strongly correlated variables, the kind of structure PCA summarises well *once* the data are stationary.)

Question 4 (15 points): the VAR

A note on the sample exam's printing: the matrices in the PDF are garbled. The most likely reading is $\delta = (1, 10)'$ and $\Theta = \begin{pmatrix} -0.1 & 0.2 \\ 0 & 0.2 \end{pmatrix}$, with uncorrelated errors. The method below works for any numbers.

(a) Impulse responses of y_t to a unit change in x_t at horizons 0 and 1.

Definition: an impulse response is the effect of a one-time unit shock to one variable's error, at time t , on a variable s periods later, holding all other shocks at zero: $\partial z_{t+s} / \partial \varepsilon_t = \Theta^s$ for a VAR(1).

Horizon 0: $\Theta^0 = I$, whose top-right element is 0. The response of y_t to a shock in x_t is **0**: y_t depends only on x_{t-1} , and the errors are uncorrelated, so y cannot react in the same period.

Horizon 1: $\Theta^1 = \Theta$; the response is the top-right element $\theta_{12} = \mathbf{0.2}$. One period after a unit shock to x , y is 0.2 higher. (Horizon 2 would give $\theta_{11}\theta_{12} + \theta_{12}\theta_{22} = 0.02$: the effect dies out.)

(b) Significant trend and stationary VAR coefficients: is cointegration appropriate?

No. Cointegration is designed for variables that are integrated (I(1)), i.e. that contain stochastic trends, and share a stationary combination. Here the VAR coefficients satisfy the stationarity condition, so shocks die out: the variables are stationary around a deterministic trend (trend-stationary), not I(1). There is no stochastic trend to share, so a cointegration model is not appropriate; the trend should simply be kept in the VAR.

(c) The curse of dimensionality.

In a VAR every variable appears with all its lags in every equation. With N variables and p lags there are $N(1 + Np) = N + pN^2$ mean parameters (plus the error covariances), so the number of parameters grows with the square of N , while the number of observations grows only with time. For example, 10 variables with one lag already require 110 parameters. With limited data the estimates become imprecise, the model overfits, and forecasts deteriorate. Remedies are parameter restrictions, penalized estimation, or reducing the dimension with factor models.

Your Formula Sheet and Final Checklist

11.1 Copy this onto your two A4 sheets

Returns. $r_t = 100(\ln P_t - \ln P_{t-1})$. Stylized facts: no autocorrelation in returns; fat tails; volatility clustering; leverage.

Stationarity. $E(y_t) = \mu$, $\text{Var}(y_t) = \sigma^2$, $\text{cov}(y_t, y_{t-k}) = \gamma_k$ for all t . ACF band $\pm 1.96/\sqrt{T}$.

AR(1) $y_t = \alpha_0 + \alpha_1 y_{t-1} + \varepsilon_t$, $|\alpha_1| < 1$, $\text{cov}(y_{t-1}, \varepsilon_t) = 0$: $\mu = \frac{\alpha_0}{1-\alpha_1}$, $\gamma_0 = \frac{\sigma^2}{1-\alpha_1^2}$, $\rho_k = \alpha_1^k$.

With $\beta_1 x_t$: $\mu = \frac{\alpha_0 + \beta_1 \mu_x}{1-\alpha_1}$; $\text{Var} = \frac{\beta_1^2 \sigma_x^2 + \sigma^2}{1-\alpha_1^2}$ (assuming $\text{cov}(y_{t-1}, x_t) = 0$).

MA(1) $y_t = \alpha_0 + \varepsilon_t + \beta_1 \varepsilon_{t-1}$: mean α_0 , variance $(1 + \beta_1^2)\sigma^2$, ACF cuts off after lag 1.

ACF/PACF: AR(p): PACF cuts off at p . MA(q): ACF cuts off at q . ARMA: both decay.

IC: $AIC = -2 \ln L + 2k$, $BIC = -2 \ln L + k \ln T$; smaller is better; BIC prefers smaller models.

Forecast: $E(y_{t+1}|I_t) = \alpha_0 + \alpha_1 y_t$; error variance $\rightarrow \gamma_0$. $MSFE = \frac{1}{T} \sum (y_f - \hat{y}_f)^2$, $MAFE = \frac{1}{T} \sum |y_f - \hat{y}_f|$.

Trends. Deterministic: $y_t = \alpha + \beta t + \varepsilon_t$ (detrend). Random walk: $y_t = y_{t-1} + \varepsilon_t$, shocks permanent, $\text{Var} = t\sigma^2$ (difference). $I(1) + I(0) = I(1)$.

Dickey-Fuller: $\Delta x_t = \alpha + \beta t + \delta x_{t-1} + \varepsilon_t$; $H_0 : \delta = 0$ (unit root), $H_1 : \delta < 0$. DF critical values (≈ -2.86 const, -3.41 trend). ADF adds $\gamma_i \Delta x_{t-i}$. KPSS: $H_0 =$ stationary. If in doubt, difference.

EMH: log prices random walk ($\beta_1 = 1$); in a returns regression $\alpha_1 = 0$, no trend.

Extend a model: dummy D_t ; mean effect $+\gamma D_t$; volatility effect $\sigma_t^2 = \exp(\omega + \lambda D_t)$; ARCH-M $+\delta \sigma_{t-1}^2$. Trend needed? t -test, AIC/BIC, MSFE, DF/plot.

VAR(1) $z_t = \delta + \Theta z_{t-1} + \varepsilon_t$. θ_{12} : y on x_{t-1} . Forecast $\hat{z}_{T+1} = \delta + \Theta z_T$. IRF = Θ^s ($\Theta^0 = I$): y to x : 0 at $s = 0$, θ_{12} at $s = 1$, $(\Theta^2)_{12}$ at $s = 2$. Parameters $N + pN^2$. Trend-stationary $\Rightarrow$ no cointegration.

ARCH(1): $\varepsilon_t = \sqrt{h_t} v_t$, $h_t = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2$; $\text{Var}(r_t|I_{t-1}) = h_t$.

GARCH(1,1): $h_t = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 h_{t-1}$; $E(h) = \frac{\alpha_0}{1-\alpha_1-\beta_1}$; short run: $\alpha_0 > 0, \alpha_1, \beta_1 \geq 0$; long run: $\alpha_1 + \beta_1 < 1$; half-life $\frac{\ln 0.5}{\ln(\alpha_1 + \beta_1)}$. Key step: $E(\varepsilon^2) = E(h)E(v^2) = E(h)$.

ARCH(2) $+\alpha_3 x_{t-1}$: $E(y) = \mu$; $E(\sigma^2) = \frac{\alpha_0 + \alpha_3 \mu_x}{1-\alpha_1-\alpha_2}$; negative $\alpha_3 \Rightarrow$ possible negative variance.

TARCH: $+\lambda_1 I\{\varepsilon < 0\} \varepsilon^2$, $\lambda_1 > 0$ leverage; $E(h) = \frac{\alpha_0}{1-\alpha_1-\beta_1-\lambda_1/2}$.

EGARCH: models $\log h_t \Rightarrow$ no positivity restrictions; leverage = coefficient on signed $\varepsilon/\sqrt{h}$ (< 0).

SV: $r_t = \varepsilon_t e^{h_t/2}$, $h_{t+1} = \mu + \phi(h_t - \mu) + \eta_t$. **HAR:** $\sigma_{t+1}^{(d)} = c + \beta^{(d)} RV^{(d)} + \beta^{(w)} RV^{(w)} + \beta^{(m)} RV^{(m)}$.

Var: $\Pr(y_t < \text{VaR}_t(\alpha)) = \alpha$; **ES** $= E(y_t | y_t < \text{VaR}_t(\alpha))$. Volatility forecasts: compare $\hat{h}$ with r^2 or RV .

Big data: factor model $x_t = \Lambda f_t + \varepsilon_t$; PCA via eigenvalues (variance explained); choose r : scree elbow, cumulative share, IC, forecasts. Ridge $+\lambda \sum \beta^2$ (shrinks, no zeros; 1 variable: $\hat{\beta} = \frac{\sum xy}{\sum x^2 + \lambda}$); Lasso $+\lambda \sum |\beta|$ (exact zeros, selects); elastic net both. λ by BIC or CV. 2^N subsets. Stationarity assumed!

Extras. $JB = \frac{T}{6} (S^2 + \frac{(K-3)^2}{4})$ (H_0 normal). Ljung-Box $Q = T(T+2) \sum_{k \leq m} \frac{\hat{\rho}_k^2}{T-k}$ (H_0 no autocorrelation). AR(1) MA(∞): $y_t = \frac{\alpha_0}{1-\alpha_1} + \sum_k \alpha_1^k \varepsilon_{t-k}$. RW with drift $= x_0 + \mu t + \sum \varepsilon$. VAR errors Σ (σ_{12}

= same-period covariance). GARCH(p, q). Normal ES(1%) $\approx -2.67\sigma$, VaR(1%) $= -2.33\sigma$. Factor regression $y_t = \tilde{\beta}' f_t$, $\tilde{\beta}' = \beta' \Lambda$. PCA: $Av = \lambda v$; $\hat{\Lambda}$ = first r eigenvectors of $X'X/T$, $\hat{f}_t = \hat{\Lambda}' x_t$. $BIC(\lambda) = \ln(\frac{1}{T} \sum e_t^2) + s(\lambda) \frac{\ln T}{T}$. glmnet: $\alpha = 1$ Lasso, 0 Ridge. OLS on AR: biased, consistent; heteroskedasticity: consistent, wrong SEs (HAC).

11.2 The 25 questions to be able to answer

If you can answer all of these in two or three sentences, you are ready.

1. What makes time series data different from cross-section data?
2. Name four stylized facts of financial returns.
3. Define stationarity. Is a trending series stationary? A seasonal one?
4. What do the ACF and PACF measure, and how do they identify AR and MA models?
5. Derive the long-run mean and variance of an AR(1), stating your assumptions.
6. Redo question 5 with an extra stationary regressor x_t .
7. What is the difference between a deterministic and a stochastic trend? What happens to a shock in each?
8. Write the Dickey–Fuller regression with constant and trend, and state H_0 and H_1 .
9. What is a spurious regression?
10. What does the efficient market hypothesis predict for a regression of returns on lagged returns and a trend?
11. Add a dummy effect to the mean of a model, and to its variance. Which parameter captures each?
12. Give four ways to decide whether a trend belongs in a model.
13. Write a two-variable VAR(1) as two equations and interpret each coefficient.
14. Define the impulse response and compute it at horizons 0, 1 and 2 for a given Θ .
15. When is a cointegration model appropriate, and when not?
16. Explain the curse of dimensionality with a parameter count.
17. Why use a conditional volatility model? Give an economic example.
18. Derive the unconditional variance of a GARCH(1,1) (or an ARCH model with an extra variable) and state the conditions.
19. Why can a negative coefficient in the variance equation be a problem, and how does EGARCH solve it?

20. Why Ridge instead of OLS when there are more variables than observations? How do you choose the number of principal components, and why must the data be stationary?
21. Why is OLS on an AR model biased but consistent? What do heteroskedasticity and autocorrelation do to OLS?
22. Why is a random walk with drift said to contain both a deterministic and a stochastic trend?
23. Give two reasons to model time series jointly in a VAR, with an example of each.
24. Compute a 1% Value-at-Risk from a volatility forecast, and explain how expected shortfall differs.
25. Explain how principal components are estimated (eigenvectors) and how λ is chosen by BIC or cross-validation.

11.3 A calm one-week plan

- Day 1** Chapters 1–3. Make sure “stationary” and “ACF” feel natural.
- Day 2** Chapter 4. Do the AR(1) mean and variance derivations twice from memory, once with x_t .
- Day 3** Chapters 5 and 6. Write the Dickey–Fuller hypotheses until automatic; practise three model extensions.
- Day 4** Chapter 7. Compute an impulse response for a new Θ of your own.
- Day 5** Chapter 8. Derive the GARCH long-run variance from memory; explain the log trick aloud.
- Day 6** Chapter 9, then write your two formula sheets (Chapter 11 is the template).
- Day 7** Do the sample exam (Chapter 10) under time pressure *before* rereading the solutions. Then answer the 25 questions.

The idea in plain words

On exam day, remember the three habits that earn most points: **write full sentences**, **show every step**, and **state your assumptions**. A correct number without explanation earns little; a clear explanation with a small slip earns a lot.

Chapter A

R Toolkit and Lessons from the Tutorials

The syllabus includes two R sessions (basics and intermediate). R code itself is *not graded*, but the exam can ask what an analysis does, and the tutorials taught a few lessons that appear nowhere else. This appendix collects both.

A.1 The commands you need, topic by topic

```
## Data and returns
d <- read.csv("TESLA.csv") # CSV files: read.csv, not read_excel
d$Date <- as.Date(d$Date, format = "%m/%d/%Y") # text -> real dates
m <- merge(tsla, jpm, by = "Date") # keep only shared dates
r <- 100 * diff(log(d$Close)) # log returns in %
plot(r, type = "l"); hist(r); acf(r); pacf(r) # first look
library(ggplot2) # presentation-quality plots
ggplot(d, aes(Date, Close)) + geom_line() + labs(y = "Price (USD)")

## ARMA
m1 <- arima(r2, order = c(1, 0, 0)) # AR(1); c(p, 0, q) in general
m1$code # 0 = estimation converged
coef(m1); AIC(m1); BIC(m1) # note: 'intercept' = mean
Box.test(residuals(m1), lag = 20, type = "Ljung-Box", fitdf = 1)
library(forecast); f <- forecast(m1, h = 150) # forecasts and intervals

## Unit roots
library(urca)
summary(ur.df(y, type = "trend", selectlags = "BIC")) # ADF with constant + trend

## VAR
library(vars)
v <- VAR(data.frame(rw, rb), p = 1, type = "const")
summary(v); VARselect(data.frame(rw, rb))$selection # lag choice
causality(v, cause = "rb") # Granger test
plot(irlf(v, n.ahead = 12)) # impulse responses

## Volatility
library(fGarch)
g <- garchFit(~ garch(1, 1), data = r, cond.dist = "norm")
summary(g); g@sigma.t # conditional volatility

## High-dimensional
library(glmnet)
cv <- cv.glmnet(X, y, alpha = 1) # alpha 1 = Lasso, 0 = Ridge
coef(cv, s = "lambda.min")
pc <- prcomp(X, scale. = TRUE); summary(pc) # PCA + variance shares
```

Practical tip: never copy code from a PDF — PDFs turn straight quotes into curly ones, and R rejects them. Use plain .R files.

A.2 Lessons from the tutorials worth remembering

- **Simulation reveals properties.** Because the truth is known, simulation shows unbiasedness (estimates average to the true value) and consistency (their spread shrinks as the sample grows) — Chapter 2.
- **The direction of returns is unpredictable too.** A series equal to 1 on up-days and 0 on down-days (Tesla: 51% up-days) shows no autocorrelation — the efficient-market message survives even this crude transformation.
- **Normalised indices compare growth.** Dividing each price by its first value (times 100) puts two stocks on the same footing: Tesla grew from 100 to about 2,000 over 2010–2020, JPMorgan to about 290 — with far higher risk for Tesla.
- **Check convergence before comparing models** — a model that did not converge can still print the “best” AIC (Chapter 4).
- **Rolling-window estimation tests parameter stability.** Re-estimate a model on a moving window (e.g. the last 500 days) and plot the coefficient over time. For JPMorgan’s squared returns the AR(1) coefficient was 0.47 over the full sample, but jumped above 1 in windows covering March 2020. *Window length is a trade-off:* a short window reacts quickly to genuine change but is noisy; a long window is smooth but slow and dilutes breaks.
- **Expanding-window forecasts are honest forecasts.** Re-estimating on all data up to each date and forecasting one step ahead uses no future information — unlike in-sample fitted values (GARCH tutorial).

Chapter B

The Group Assignment in Brief

The final group assignment is 30% of the grade: research proposal (5%), report (15%), presentation with individual questions (10%). Each part needs at least a 5.5 overall assignment grade.

Proposal (presented 24 September). It must contain: the research question and its economic motivation; the main hypothesis or hypotheses; the data; the proposed econometric method; the main econometric or identification concern; and what the analysis will contribute.

Report (6–8 pages, due 15 October, 23:00, with your code). Required structure: research question and motivation; data and construction choices; methodology (explained *and motivated*); results in tables and figures; **economic interpretation** (magnitude and economic significance, not only statistical significance); robustness and limitations; conclusion (what can and cannot be concluded); division of labour.

Presentation (15 minutes including discussion), followed by one or two individual questions on *any* part of the project. Every member must understand the whole analysis. A penalty applies for submission mistakes (wrong file format, missing team information).

The idea in plain words

The grading principle, straight from the syllabus: **a simple method that is appropriately chosen and correctly applied is better than a sophisticated method that does not fit the question.** Appendix C's method map helps you choose.

Chapter C

Coverage Map: Every Syllabus Item, Every Slide Topic

C.1 Which model for which problem?

The syllabus's learning objectives ask you to *evaluate which model fits a financial problem* and *translate a problem into a standard model*. This table is that translation:

If the question is about...	use	watch out for
the dynamics of one stationary series; short-term forecasts	ARMA (Ch. 4)	ACF/PACF, convergence, overfitting
whether a series is stationary; prices vs returns; regressions in levels	ADF/KPSS (Ch. 5)	H_0 = unit root; include trend; spurious regression
calendar or event effects	dummies (Ch. 6)	mean or variance? dummy trap
several series influencing each other; lead-lag; shock transmission	VAR, Granger, IRF (Ch. 7)	stationary inputs; parameter count
risk, volatility clustering, VaR	ARCH/GARCH family (Ch. 8)	positivity; persistence; leverage
many predictors for one target	Ridge/Lasso (Ch. 9)	choosing λ ; stationarity
many series moving together	factors/PCA (Ch. 9)	number of components; stationarity

C.2 The syllabus schedule

1 Sep, lecture	Introduction → Chapters 1–3.
3 Sep, tutorial	R programming basics → Appendix A.
8 Sep, tutorial	ARMA models → Chapter 4.
10 Sep, tutorial	Unit root testing → Chapter 5.
15/17 Sep	VAR models → Chapter 7.
22 Sep, tutorial	R programming intermediate → Appendix A.
24 Sep	Research proposal → Appendix B.
29 Sep/1 Oct	Conditional volatility → Chapter 8 (and Chapter 6 for extensions).
6/8 Oct	Penalized regression → Chapter 9.
13 Oct	Principal components → Chapter 9.
15 Oct	Final presentation → Appendix B.

C.3 Slide topics, deck by deck

Introduction

time series definition, stock and flow data (Ch. 1); uses of time series and uncertainty (Ch. 1); special features: time dependence, forecasting (Ch. 1); stylized facts: trends, meandering series, changing volatility, correlated series (Ch. 1); descriptive statistics, Jarque–Bera (Ch. 1); OLS assumptions and time dependence (Ch. 2); seasonality and seasonal spurious correlation (Ch. 1, 2); trends vs stationarity, strict and covariance stationarity, existence of moments (Ch. 2); trend-induced spurious regression (Ch. 2).

ARMA

the “no trends” condition (Ch. 4); $AR(p)$ and its stationarity condition (Ch. 4); long-run mean (short route and $MA(\infty)$ route) and variance (Ch. 4); ACF/PACF identification and information criteria (Ch. 4); $MA(q)$, its intuition and estimation issue (Ch. 4); ARMA and the abbreviations ARMAX, ARIMA (Ch. 4); maximum likelihood (Ch. 4); misspecification and remedies (Ch. 4); forecasting, forecast errors and their variance, horizons, MSFE/MAFE (Ch. 4).

Unit roots

trend detection is hard; dangers of ignoring trends (cheese, unemployment nowcasting); spurious regression; random walks, the drunkard’s walk, nonlinear-looking random walks; deterministic trends; combining trends (drift); effect of shocks and impulse response ρ^h ; mean reversion; removing trends; forecasting with linear and stochastic trends; Dickey–Fuller, the four-outcome table, ADF, KPSS; rules of thumb 1–5 (all in Chapter 5).

VAR

relation to AR models; reasons to model jointly; system and vector notation; error covariance Σ ; stationarity (inverse roots); structural VAR; model selection; misspecification remedies; forecasting; impulse responses Θ^s (all in Chapter 7); Granger causality and curse of dimensionality (Chapters 7 and 10).

Conditional volatility

time-varying volatility, OLS problems and HAC; volatility clustering; conditional volatility; VaR, expected shortfall, leverage; measuring volatility, realized volatility; HAR model and its derivation; ARCH(1); GARCH(1,1), long-run variance, conditions; GARCH(p, q); TARARCH and its long-run variance; EGARCH; stochastic volatility; forecast evaluation of means and volatilities (all in Chapter 8); dummies and exogenous variables in the variance, ARCH-in-mean (Chapters 6 and 8).

High-dimensional

high-dimensional data and the OLS problem (simulation); aggregation vs subset selection; stationarity assumed; factor models, observed factors (Fama–French), unobserved factors; eigenvalues and eigenvectors; estimation by principal components; properties and interpretation; number of factors; factor forecasts; subset selection and 2^N ; penalized regression, L_q norms and Lagrangian duality; Ridge (with derivation) and Lasso; elastic net; glmnet; choosing λ by BIC and cross-validation; forecasting with penalized regression (all in Chapter 9).